

## Answers

|        |       |        |        |       |       |       |        |
|--------|-------|--------|--------|-------|-------|-------|--------|
| 1.C    | 2.A   | 3.B    | 4.D    | 5.C   | 6.D   | 7.C   | 8.C    |
| 9.D    | 10.B  | 11.60  | 12.170 | 13.A  | 14.A  | 15.A  | 16.C   |
| 17.A   | 18.C  | 19.2   | 20.A   | 21.C  | 22.4  | 23.A  | 24.50  |
| 25.C   | 26.C  | 27.B   | 28.A   | 29.D  | 30.84 | 31.64 | 32.325 |
| 33.340 | 34.10 | 35.150 | 36.D   | 37.B  | 38.A  | 39.B  | 40.40  |
| 41.D   | 42.C  | 43.50  | 44.A   | 45.20 | 46.A  | 47.C  | 48.42  |

## Explanations

### 1. C

No fishes breathe through lungs but all whales breathe through lungs. So, no whale can be a fish. Therefore, ADE is a valid combination of logically connected statements.  
Hence, option C is the correct answer.

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### 2. A

In option A, D can be concluded from C and E.  
In option B, A, C and B are not connected.  
In option C, C cannot be inferred from B and D.  
In option D, E and F are not related.

Hence, option A is the correct answer.

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### 3. B

Everyone who is sane can do logic and all who can do logic are fit to serve on a jury. So, all who can do logic are fit to serve on a jury.  
Hence, option B is the correct answer.

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### 4. D

In option A, E is not a valid conclusion from A and B.

In option B, F cannot be inferred from A and C.

In option C, E cannot be concluded from A and D.

In option D, If some learned men are candid but all candid men recognize merit in a rival, it can be concluded that some men recognize merit in a rival are learned.

Hence, option D is the correct answer.

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### 5. C

In option A, A is inconsistent with E. It is not given in A that only those who achieve great ends are happy.

In option B, F contradicts with B.

In option C, B can be inferred from A and C.

In option D, F cannot be concluded from A and D.

Hence, option C is the correct answer.

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## 100+ CAT VARC Questions



6. D

Nothing uncertain is worth dying for. and all beliefs are uncertain. So, no belief is worth dying for. Hence, option D is the correct answer.

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7. C

There are six sentences given. So, there can be many combinations of the sentences possible. Therefore, we will check each of the option to find the correct answer.

In option A, BCD is given. Statement C says all of my closest friends disapprove of me and D says all who disapprove of me are aardvarks. So, all my closest friends must be aardvarks. But, this contradicts with statement B. Thus, option A is incorrect.

In option B, statement D cannot be inferred from A and B.

In option C, all three statements are logically connected.

In option D, E cannot be inferred from A and B.

Hence, option C is the correct answer.

[VIEW SOLUTION](#)

8. C

All students of this college rank as University students and all University students are eligible to enter for the prize. Therefore, all first year students are entitled to enter for the prize.

Hence, option C is the correct answer.

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9. D

ADF is false as we cannot infer F(Some mammals are whales.) even if A(Some mammals are carnivores.) and D(All whales are carnivores.) are true.

ABC and AEF all statements are different, the set is not related.

All whales are mammals and all whales are aquatic animals. So, some aquatic animals are mammals. Therefore, BCE is valid combination.

Hence, option D is the correct answer.

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10. B

Some men are men of artistic genius and all men are men of scientific ability. Therefore, all men of artistic genius are men of scientific ability.

Thus, ACE is the correct combination.

Hence, option B is the correct answer.

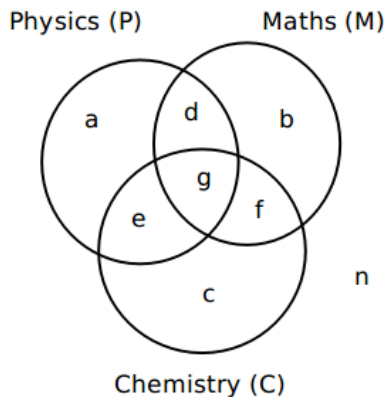
## CAT Expected Questions PDF

cracku



11.60

It is given that 200 candidates scored above 90th percentile overall in CET. Let the following Venn diagram represent the number of persons who scored above 80 percentile in CET in each of the three sections:



From (1),  $n = 0$

From (2),  $d + e + f = 150$

From (3),  $a = b = c$

Since, there are a total of 200 candidates

$$a + b + c + g = 200 - 150 = 50$$

$$3a + g = 50 \Rightarrow a < 17$$

$$\text{From (4), } (b + f + c) : (a + d + b) : (a + e + c) = 4 : 2 : 1$$

$$\text{Or } (2a + f) : (2a + d) : (2a + e) = 4 : 2 : 1$$

$$\text{or, } 6a + (d + e + f) = 7x$$

$$\text{Or, } 6a + 150 = 7x$$

So,  $a$  can be 3 or 10

$x$  can be 24 or 30

$$2a + e \text{ can be } 24 \text{ or } 30 \Rightarrow e \text{ can be } 18 \text{ or } 10$$

$$2a + d \text{ can be } 48 \text{ or } 60 \Rightarrow d \text{ can be } 42 \text{ or } 40$$

$$2a + f \text{ can be } 96 \text{ or } 120 \Rightarrow f \text{ can be } 90 \text{ or } 100$$

$$3a + g = 50 \Rightarrow g \text{ can be } 41 \text{ or } 20$$

Among the candidates who are at or above 90th percentile, the candidates who are at or above 80th percentile in at least two sections are selected for AET. Hence, the candidates represented by  $d, e, f$  and  $g$  are selected for AET.

BIE will consider the candidates who are appearing for AET and are at or above 80th percentile in P. Hence, BIE will consider the candidates represented by  $d, e$  and  $g$ , which can be 104 or 80.

BIE will conduct a separate test for the other students who are at or above 80th percentile in P. Given that there are a total of 400 candidates at or above 80th percentile in P, and since there are 104 or 80 candidates at or above 80th percentile in P and are at or above 90th percentile overall, there must be 296 or 320 candidates at or above 80th percentile in P who scored less than 90th percentile overall.

From the given condition,  $g$  is a multiple of 5.

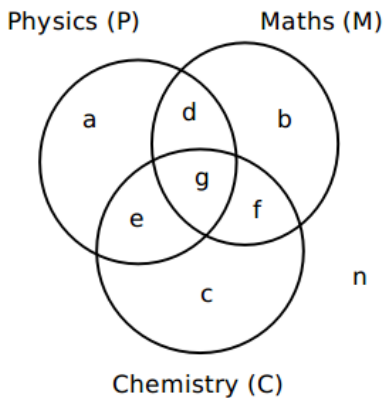
So,  $g = 20$ .

The number of candidates at or above 90th percentile overall and at or above 80th percentile in both P and M =  $d + g = 60$ .

Hence, 60 is the correct answer.

12.170

It is given that 200 candidates scored above 90th percentile overall in CET. Let the following Venn diagram represent the number of persons who scored above 80 percentile in CET in each of the three sections:



From (1),  $n = 0$

From (2),  $d + e + f = 150$

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Since, there are a total of 200 candidates

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$$\text{or, } 6a + (d + e + f) = 7x$$

$$\text{Or, } 6a + 150 = 7x$$

So,  $a$  can be 3 or 10

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$$2a + e \text{ can be } 24 \text{ or } 30 \Rightarrow e \text{ can be } 18 \text{ or } 10$$

$$2a + d \text{ can be } 48 \text{ or } 60 \Rightarrow d \text{ can be } 42 \text{ or } 40$$

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Among the candidates who are at or above 90th percentile, the candidates who are at or above 80th percentile in at least two sections are selected for AET. Hence, the candidates represented by  $d, e, f$  and  $g$  are selected for AET.

BIE will consider the candidates who are appearing for AET and are at or above 80th percentile in P. Hence, BIE will consider the candidates represented by  $d, e$  and  $g$ , which can be 104 or 80.

BIE will conduct a separate test for the other students who are at or above 80th percentile in P. Given that there are a total of 400 candidates at or above 80th percentile in P, and since there are 104 or 80 candidates at or above 80th percentile in P and are at or above 90th percentile overall, there must be 296 or 320 candidates at or above 80th percentile in P who scored less than 90th percentile overall.

In this question,  $g = 20$ .

Number of candidates shortlisted for AET

$$= d + e + f + g$$

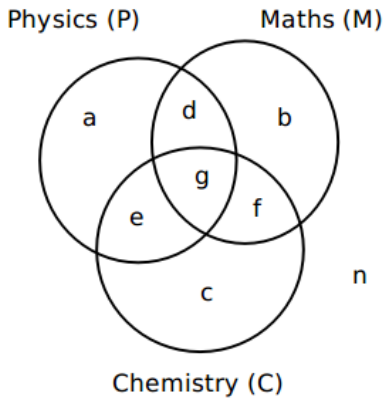
$$= 40 + 10 + 100 + 20$$

$$= 170$$

Hence, 170 is the correct answer.

13.A

It is given that 200 candidates scored above 90th percentile overall in CET. Let the following Venn diagram represent the number of persons who scored above 80 percentile in CET in each of the three sections:



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$$2a + d \text{ can be } 48 \text{ or } 60 \Rightarrow d \text{ can be } 42 \text{ or } 40$$

$$2a + f \text{ can be } 96 \text{ or } 120 \Rightarrow f \text{ can be } 90 \text{ or } 100$$

$$3a + g = 50 \Rightarrow g \text{ can be } 41 \text{ or } 20$$

Among the candidates who are at or above 90th percentile, the candidates who are at or above 80th percentile in at least two sections are selected for AET. Hence, the candidates represented by  $d$ ,  $e$ ,  $f$  and  $g$  are selected for AET.

BIE will consider the candidates who are appearing for AET and are at or above 80th percentile in P. Hence, BIE will consider the candidates represented by  $d$ ,  $e$  and  $g$ , which can be 104 or 80.

BIE will conduct a separate test for the other students who are at or above 80th percentile in P. Given that there are a total of 400 candidates at or above 80th percentile in P, and since there are 104 or 80 candidates at or above 80th percentile in P and are at or above 90th percentile overall, there must be 296 or 320 candidates at or above 80th percentile in P who scored less than 90th percentile overall.

From the given condition, the number of candidates at or above 90th percentile overall and at or above 80th percentile in P in CET =  $(3 + 18 + 42 + 41) = 104$ .

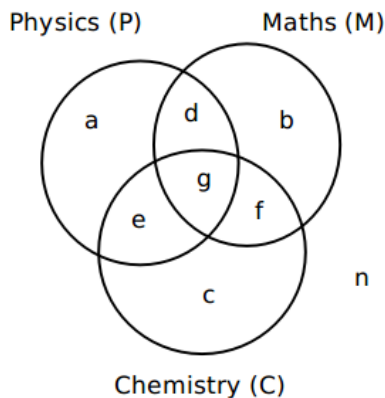
The number of candidates who have to sit for separate test =  $(400 - 104 + 3) = 296 + 3 = 299$  (we have added 3 for those who have scored more than 80th percentile only in P which is ' $a$ ').

Hence, option A is the correct answer.

[▶ VIDEO SOLUTION](#)

14. A

It is given that 200 candidates scored above 90th percentile overall in CET. Let the following Venn diagram represent the number of persons who scored above 80 percentile in CET in each of the three sections:



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From (2),  $d + e + f = 150$

From (3),  $a = b = c$

Since, there are a total of 200 candidates

$$a + b + c + g = 200 - 150 = 50$$

$$3a + g = 50 \Rightarrow a < 17$$

$$\text{From (4), } (b + f + c) : (a + d + b) : (a + e + c) = 4 : 2 : 1$$

$$\text{Or } (2a + f) : (2a + d) : (2a + e) = 4 : 2 : 1$$

$$\text{or, } 6a + (d + e + f) = 7x$$

$$\text{Or, } 6a + 150 = 7x$$

So,  $a$  can be 3 or 10

$x$  can be 24 or 30

$$2a + e \text{ can be } 24 \text{ or } 30 \Rightarrow e \text{ can be } 18 \text{ or } 10$$

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Among the candidates who are at or above 90th percentile, the candidates who are at or above 80th percentile in at least two sections are selected for AET. Hence, the candidates represented by  $d, e, f$  and  $g$  are selected for AET.

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BIE will conduct a separate test for the other students who are at or above 80th percentile in P. Given that there are a total of 400 candidates at or above 80th percentile in P, and since there are 104 or 80 candidates at or above 80th percentile in P and are at or above 90th percentile overall, there must be 296 or 320 candidates at or above 80th percentile in P who scored less than 90th percentile overall.

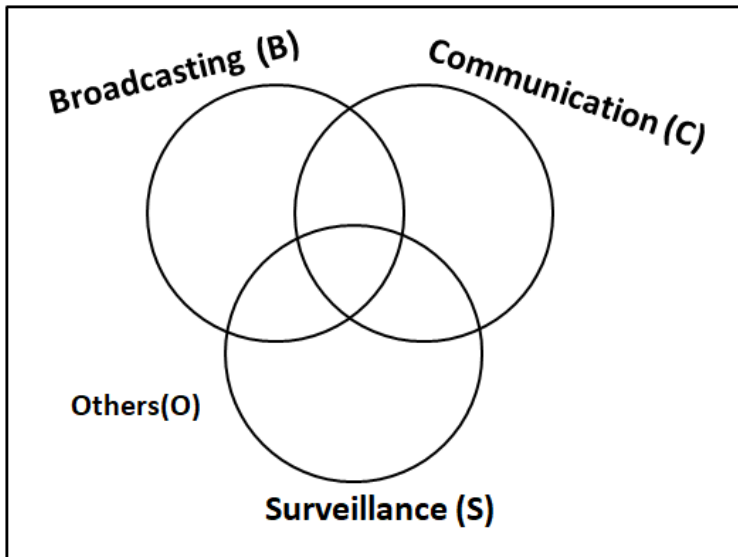
$a$  can be 3 or 10

Hence, Option A is the correct answer.

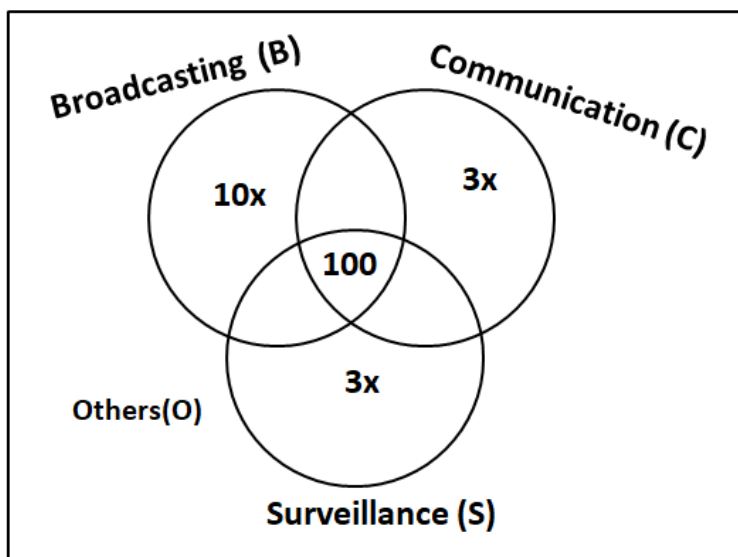
VIDEO SOLUTION

**15.A**

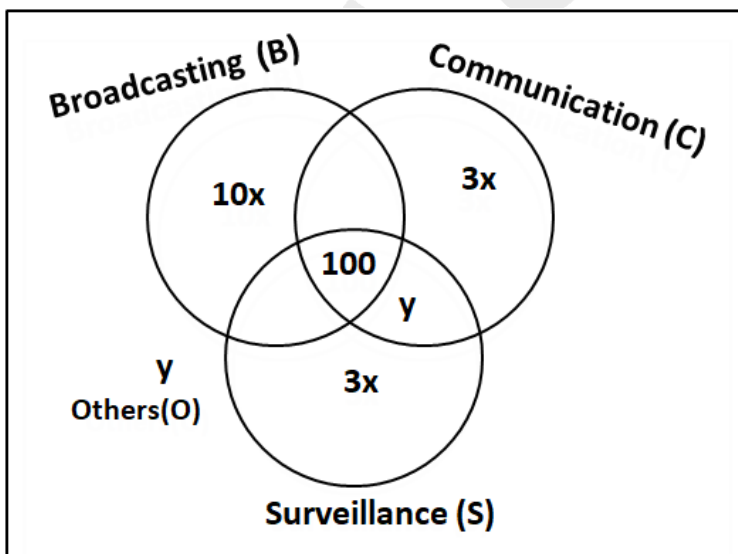
It is given that a satellite serving either B, or C, or S does not serve O. So we can say that it's basically 3 satellites broadcasting (B), communication (C), surveillance (S) which can have intersections. Those satellites which are not part of any category are placed in others. We can draw the Venn diagram as follows.



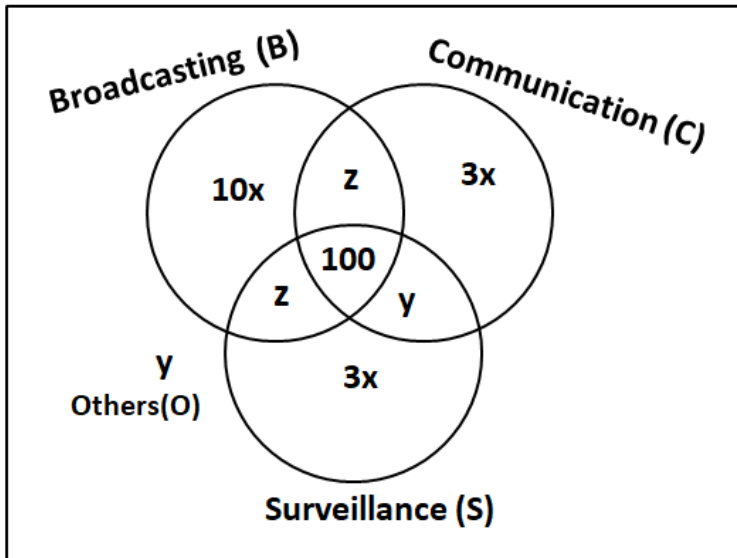
1. The numbers of satellites serving B, C, and S (though may be not exclusively) are in the ratio 2:1:1.
  2. The number of satellites serving all three of B, C, and S is 100.
  3. The number of satellites exclusively serving C is the same as the number of satellites exclusively serving S. This number is 30% of the number of satellites exclusively serving B.
  4. The number of satellites serving O is the same as the number of satellites serving both C and S but not B.
- Let '10x' be the number of satellites exclusively serving B. Then, the number of satellites exclusively serving C and S =  $0.30 \times 10x = 3x$



Let 'y' be the number of satellites serving others(O).



Let 'z' be the number of satellites serving B, C but not S. Since the numbers of satellites serving B, C, and S (though may be not exclusively) are in the ratio 2:1:1. Therefore, we can say that number of satellites serving B, S but not C = z.



It is given that

$$\Rightarrow 10x + 2z + 2y + 6x + 100 = 1600$$

$$\Rightarrow 8x + z + y = 750 \dots (1)$$

The numbers of satellites serving B, C, and S (though maybe not exclusively) are in the ratio 2:1:1.

$$\Rightarrow \frac{10x + 2z + 100}{z + 100 + 3x + y} = \frac{2}{1}$$

$$\Rightarrow 10x + 2z + 100 = 2(z + 100 + 3x + y)$$

$$\Rightarrow 4x = 100 + 2y$$

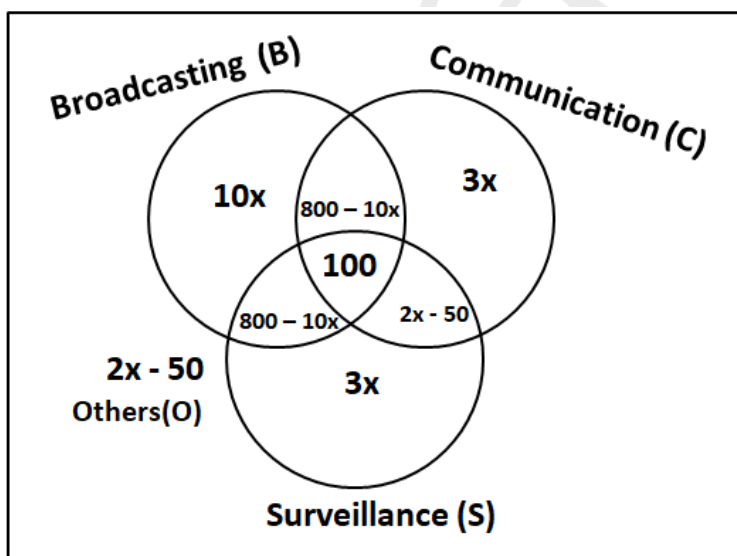
$$\Rightarrow 2x = 50 + y$$

$$\Rightarrow y = 2x - 50 \dots (2)$$

We can substitute this in equation (1)

$$\Rightarrow 8x + z + 2x - 50 = 750$$

$$\Rightarrow z = 800 - 10x \dots (3)$$



Let us define boundary condition for x,

$$\Rightarrow 2x - 50 \geq 0$$



$$\Rightarrow x \geq 25$$

$$\text{Also, } 800 - 10x \geq 0$$

$$\Rightarrow x \leq 80$$

Therefore, we can say that  $x \in [25, 80]$ .

It is given that at least 100 of the 1600 satellites were serving O.

$$\Rightarrow 2x - 50 \geq 100$$

$$\Rightarrow x \geq 75$$

The number of satellites serving S =  $100 + 800 - 10x + 2x - 50 + 3x = 850 - 5x$

At  $x_{min} = 75$ , the number of satellites serving S =  $850 - 5 \cdot 75 = 475$

At  $x_{max} = 80$ , the number of satellites serving S =  $850 - 5 \cdot 80 = 450$

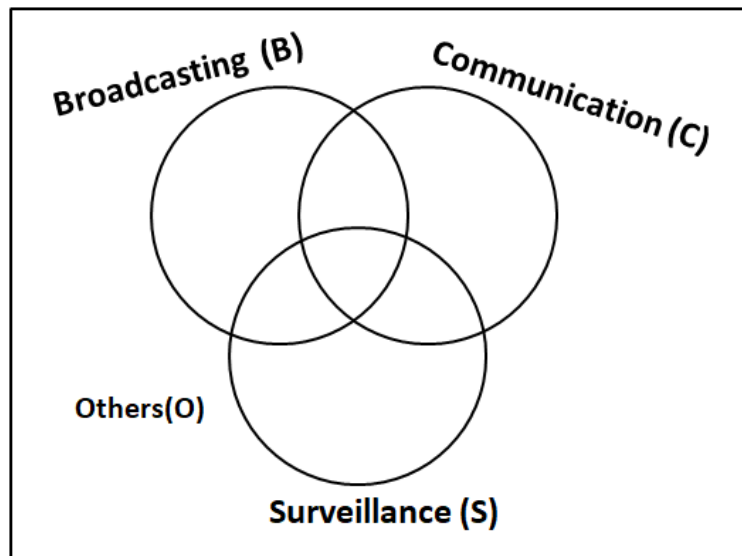
Hence, we can say that the number of satellites serving S must be from 425 to 475. Therefore, we can say that option A is the correct answer.

 VIDEO SOLUTION



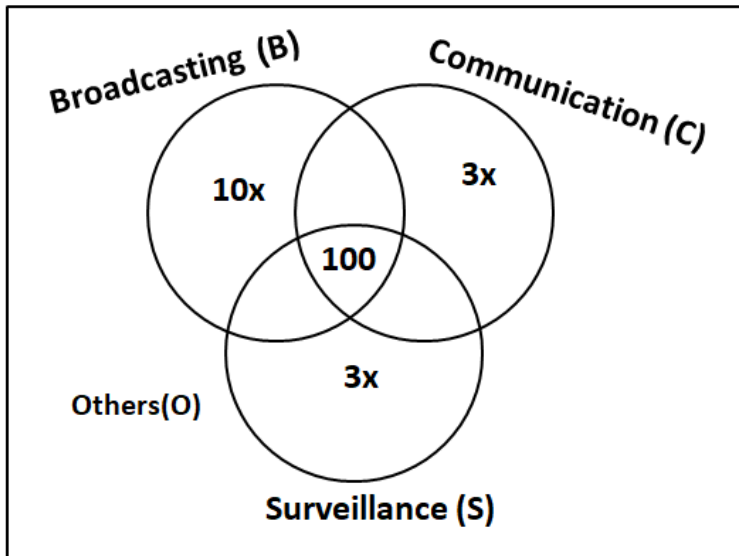
16. C

It is given that a satellite serving either B, or C, or S does not serve O. So we can say that it's basically 3 satellites broadcasting (B), communication (C), surveillance (S) which can have intersections. Those satellites which are not part of any category are placed in others. We can draw the Venn diagram as follows.

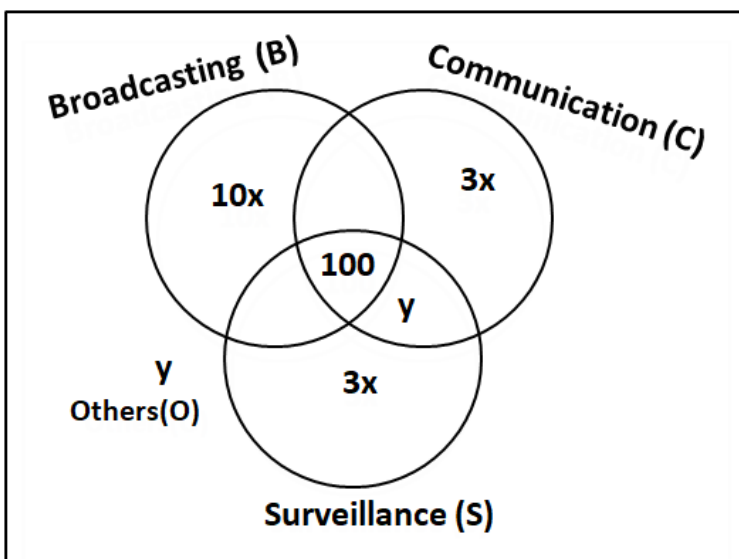


1. The numbers of satellites serving B, C, and S (though may be not exclusively) are in the ratio 2:1:1.
2. The number of satellites serving all three of B, C, and S is 100.
3. The number of satellites exclusively serving C is the same as the number of satellites exclusively serving S. This number is 30% of the number of satellites exclusively serving B.
4. The number of satellites serving O is the same as the number of satellites serving both C and S but not B.

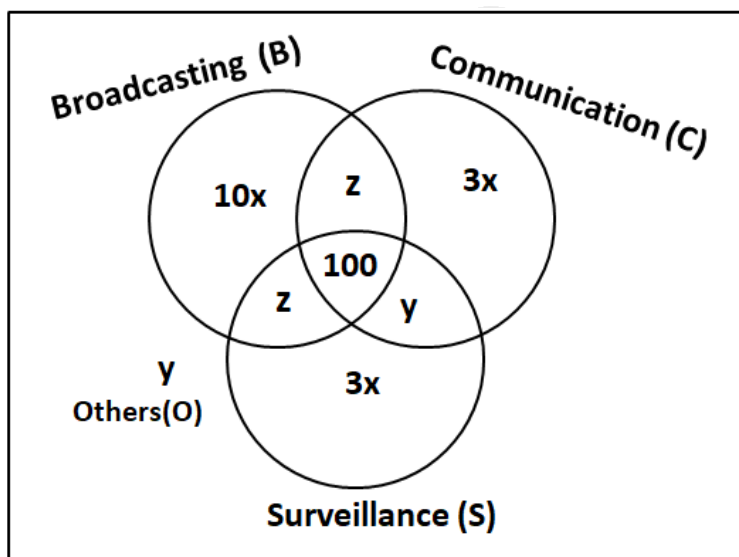
Let '10x' be the number of satellites exclusively serving B. Then, the number of satellites exclusively serving C and S =  $0.30 \cdot 10x = 3x$



Let 'y' be the number of satellites serving others(O).



Let 'z' be the number of satellites serving B, C but not S. Since the numbers of satellites serving B, C, and S (though may be not exclusively) are in the ratio 2:1:1. Therefore, we can say that number of satellites serving B, S but not C = z.



It is given that

$$\Rightarrow 10x + 2z + 2y + 6x + 100 = 1600$$

$$\Rightarrow 8x + z + y = 750 \dots (1)$$

The numbers of satellites serving B, C, and S (though maybe not exclusively) are in the ratio 2:1:1.

$$\Rightarrow \frac{10x + 2z + 100}{z + 100 + 3x + y} = \frac{2}{1}$$

$$\Rightarrow 10x + 2z + 100 = 2(z + 100 + 3x + y)$$

$$\Rightarrow 4x = 100 + 2y$$

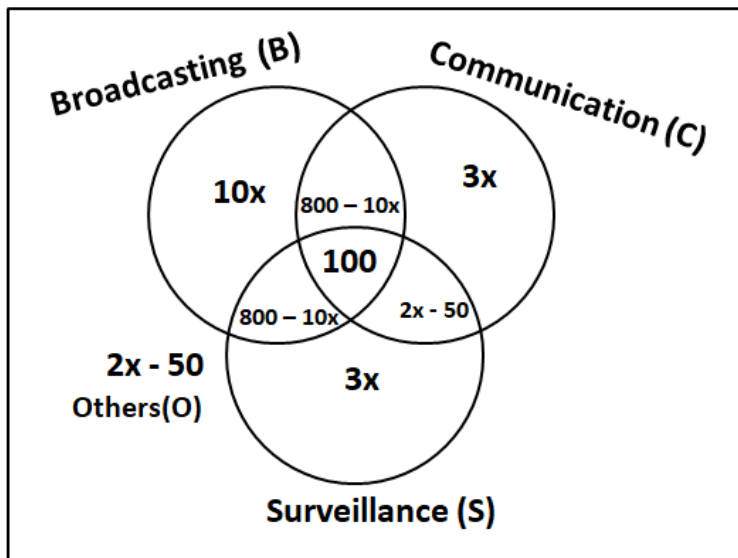
$$\Rightarrow 2x = 50 + y$$

$$\Rightarrow y = 2x - 50 \dots (2)$$

We can substitute this in equation (1)

$$\Rightarrow 8x + z + 2x - 50 = 750$$

$$\Rightarrow z = 800 - 10x \dots (3)$$



Let us define boundary condition for x,

$$\Rightarrow 2x - 50 \geq 0$$

$$\Rightarrow x \geq 25$$

$$\text{Also, } 800 - 10x \geq 0$$

$$\Rightarrow x \leq 80$$

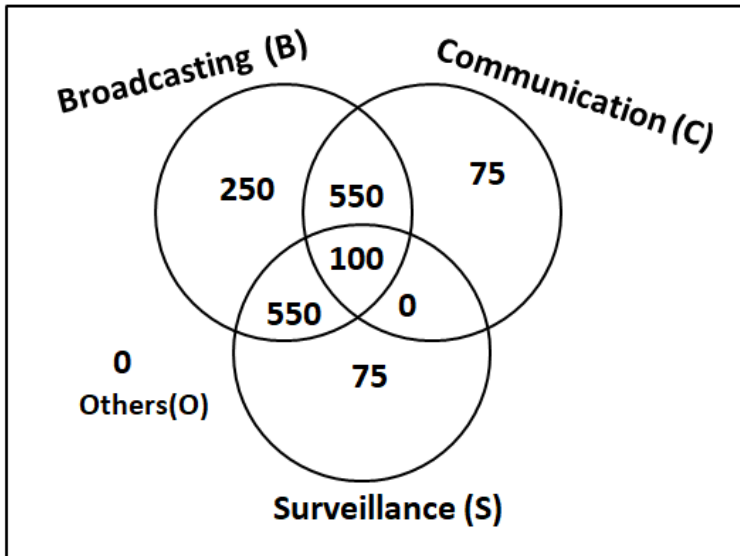
Therefore, we can say that  $x \in [25, 80]$ .

It is given that the number of satellites serving at least two among B, C, and S is 1200.

$$\Rightarrow 800 - 10x + 800 - 10x + 2x - 50 + 100 = 1200$$

$$\Rightarrow 18x = 450$$

$$\Rightarrow x = 25$$

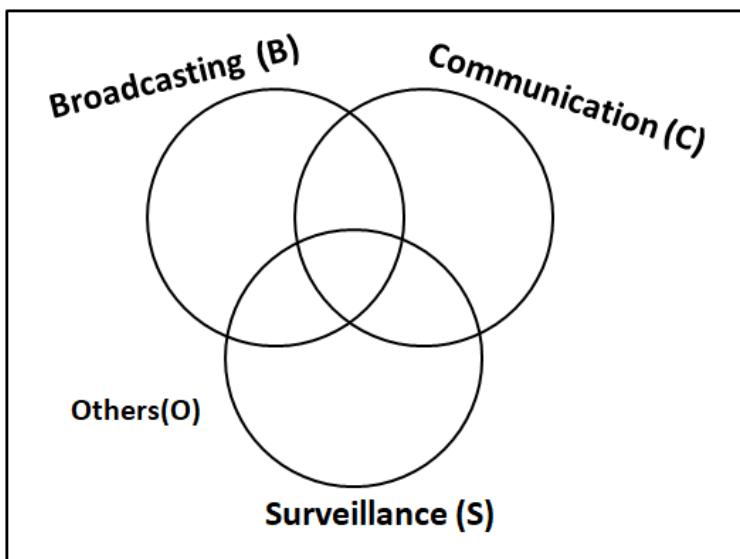


We can determine number of satellites in each of the following category. Hence, option C is definitely false. Therefore, we can say that option C is incorrect.

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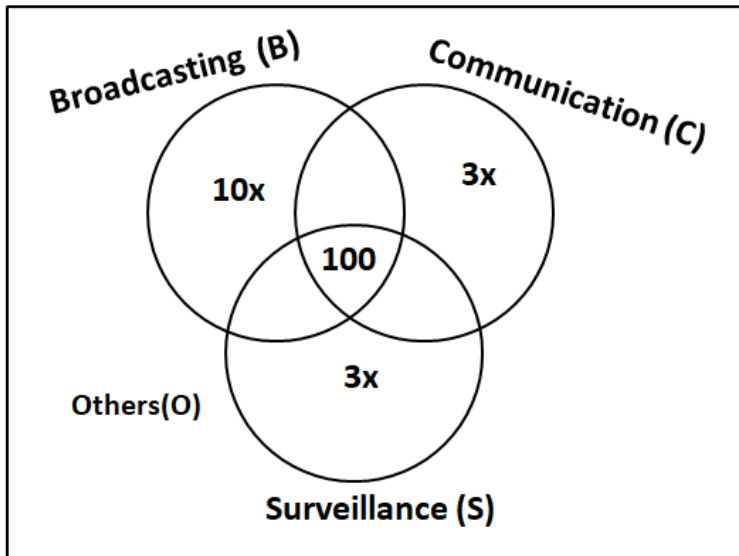
17.A

It is given that a satellite serving either B, or C, or S does not serve O. So we can say that it's basically 3 satellites broadcasting (B), communication (C), surveillance (S) which can have intersections. Those satellites which are not part of any category are placed in others. We can draw the Venn diagram as follows.

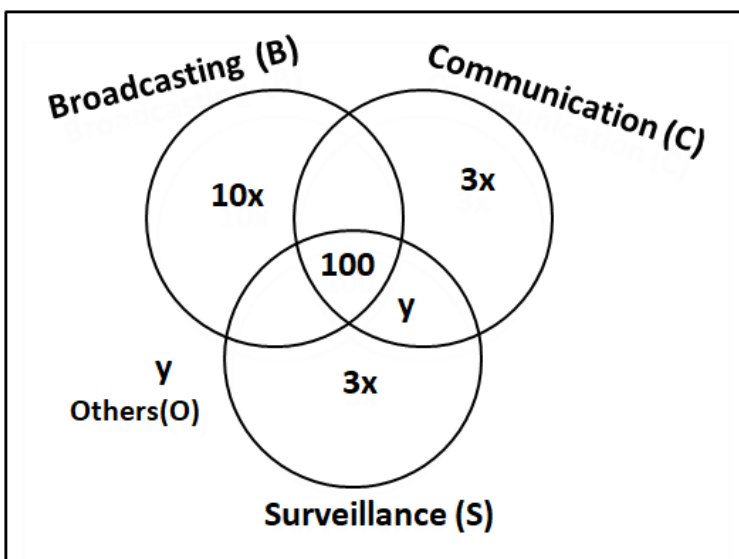


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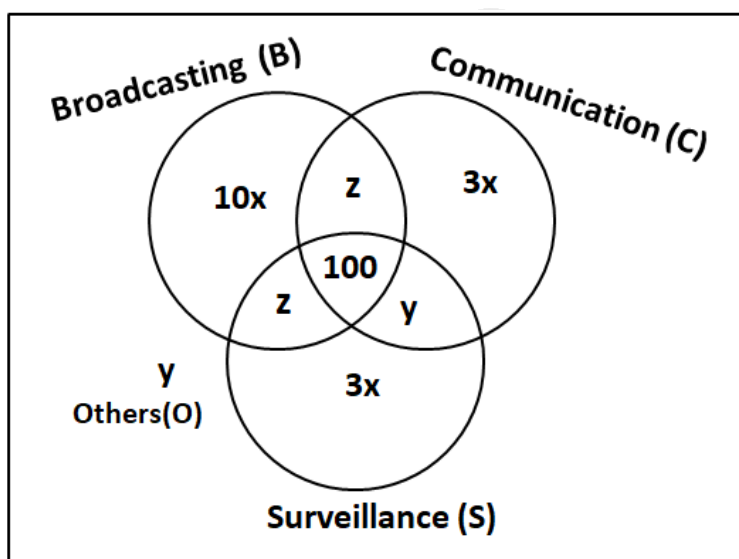
Let '10x' be the number of satellites exclusively serving B. Then, the number of satellites exclusively serving C and S =  $0.30 \times 10x = 3x$



Let 'y' be the number of satellites serving others(O).



Let 'z' be the number of satellites serving B, C but not S. Since the numbers of satellites serving B, C, and S (though may be not exclusively) are in the ratio 2:1:1. Therefore, we can say that number of satellites serving B, S but not C = z.



It is given that

$$\Rightarrow 10x + 2z + 2y + 6x + 100 = 1600$$

$$\Rightarrow 8x + z + y = 750 \dots (1)$$

The numbers of satellites serving B, C, and S (though maybe not exclusively) are in the ratio 2:1:1.

$$\Rightarrow \frac{10x + 2z + 100}{z + 100 + 3x + y} = \frac{2}{1}$$

$$\Rightarrow 10x + 2z + 100 = 2(z + 100 + 3x + y)$$

$$\Rightarrow 4x = 100 + 2y$$

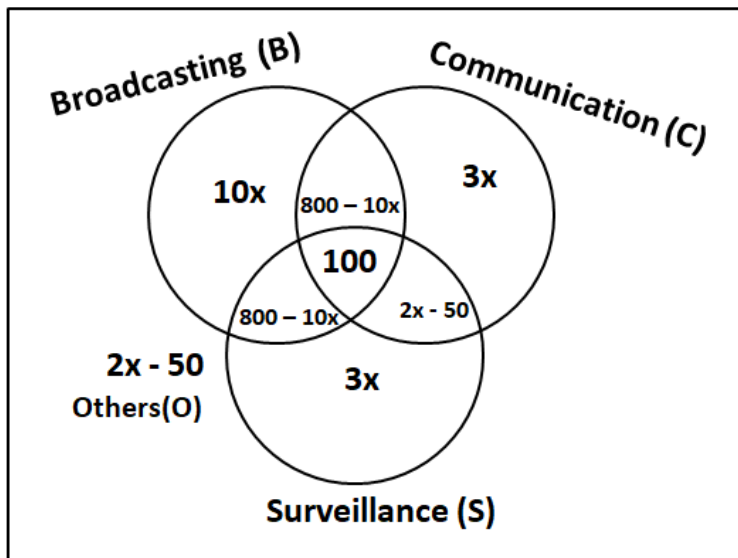
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$$\Rightarrow 8x + z + 2x - 50 = 750$$

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Let us define boundary condition for x,

$$\Rightarrow 2x - 50 \geq 0$$

$$\Rightarrow x \geq 25$$

$$\text{Also, } 800 - 10x \geq 0$$

$$\Rightarrow x \leq 80$$

Therefore, we can say that  $x \in [25, 80]$ .

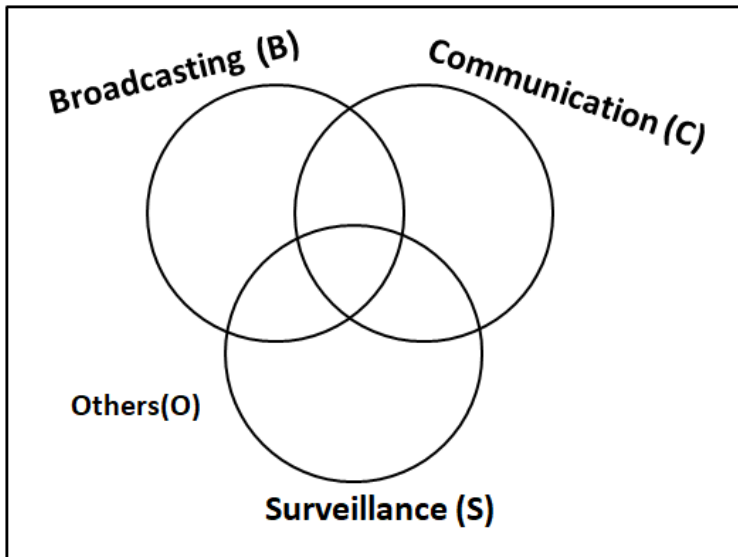
The number of satellites serving B exclusively =  $10x$ . This will be the minimum when 'x' is the minimum.

At  $x_{min} = 25$ , The number of satellites serving B exclusively =  $10 \times 25 = 250$ . Hence, option A is the correct answer.

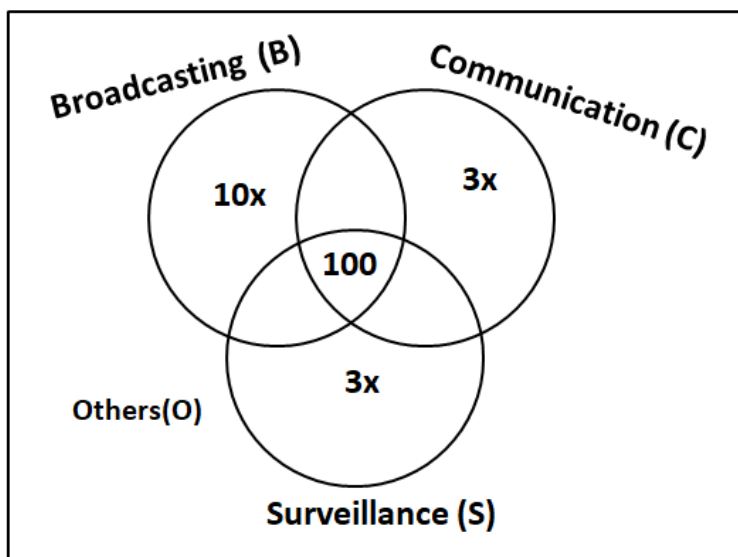
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**18.C**

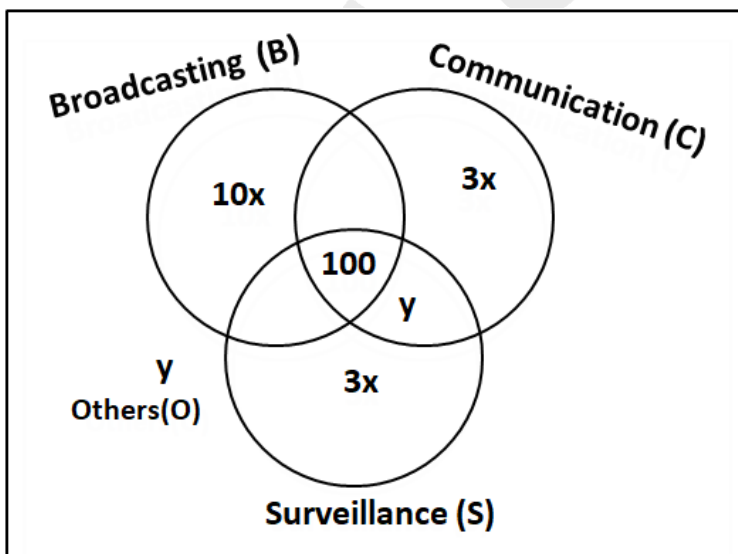
It is given that a satellite serving either B, or C, or S does not serve O. So we can say that it's basically 3 satellites broadcasting (B), communication (C), surveillance (S) which can have intersections. Those satellites which are not part of any category are placed in others. We can draw the Venn diagram as follows.



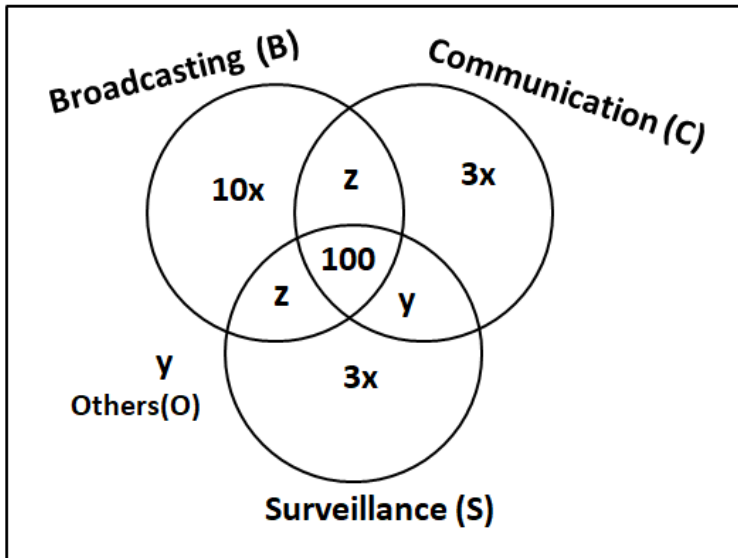
1. The numbers of satellites serving B, C, and S (though may be not exclusively) are in the ratio 2:1:1.
  2. The number of satellites serving all three of B, C, and S is 100.
  3. The number of satellites exclusively serving C is the same as the number of satellites exclusively serving S. This number is 30% of the number of satellites exclusively serving B.
  4. The number of satellites serving O is the same as the number of satellites serving both C and S but not B.
- Let '10x' be the number of satellites exclusively serving B. Then, the number of satellites exclusively serving C and S =  $0.30 \times 10x = 3x$



Let 'y' be the number of satellites serving others(O).



Let 'z' be the number of satellites serving B, C but not S. Since the numbers of satellites serving B, C, and S (though may be not exclusively) are in the ratio 2:1:1. Therefore, we can say that number of satellites serving B, S but not C = z.



It is given that

$$\Rightarrow 10x + 2z + 2y + 6x + 100 = 1600$$

$$\Rightarrow 8x + z + y = 750 \dots (1)$$

The numbers of satellites serving B, C, and S (though maybe not exclusively) are in the ratio 2:1:1.

$$\Rightarrow \frac{10x + 2z + 100}{z + 100 + 3x + y} = \frac{2}{1}$$

$$\Rightarrow 10x + 2z + 100 = 2(z + 100 + 3x + y)$$

$$\Rightarrow 4x = 100 + 2y$$

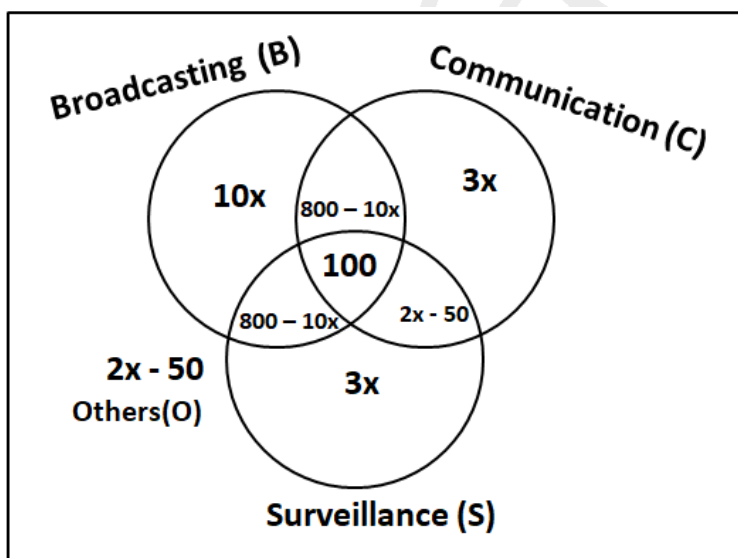
$$\Rightarrow 2x = 50 + y$$

$$\Rightarrow y = 2x - 50 \dots (2)$$

We can substitute this in equation (1)

$$\Rightarrow 8x + z + 2x - 50 = 750$$

$$\Rightarrow z = 800 - 10x \dots (3)$$



Let us define boundary condition for x,

$$\Rightarrow 2x - 50 \geq 0$$



$$\Rightarrow x \geq 25$$

$$\text{Also, } 800 - 10x \geq 0$$

$$\Rightarrow x \leq 80$$

Therefore, we can say that  $x \in [25, 80]$ .

$$\text{The number of satellites serving C} = 800 - 10x + 100 + 3x + 2x - 50 = 850 - 5x$$

$$\text{At } x = 25, \text{ The number of satellites serving C} = 850 - 5x = 850 - 5 \cdot 25 = 725$$

$$\text{At } x = 80, \text{ The number of satellites serving C} = 850 - 5x = 850 - 5 \cdot 80 = 450$$

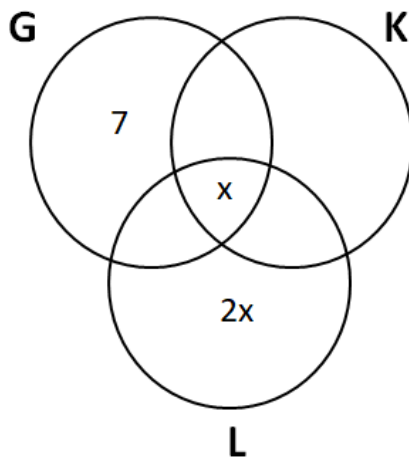
Hence, we can say that the number of satellites serving C must be between 450 and 725. Hence, option C is the correct answer.

[VIDEO SOLUTION](#)

## 19.2

Let 'x' be the number of students enrolled in all three sports. Then the number of students enrolled only in L = 2x

It is given that there are a total of 17 students enrolled in G. Also, ten students enrolled in G are also enrolled in at least one more sport. Hence, the number of students enrolled in only G = 17 - 10 = 7



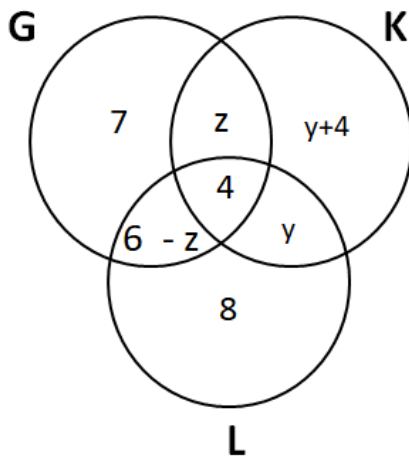
The number of students enrolled only in G is one less than the number of students enrolled only in L. Hence, the number of students enrolled only in L = 7+1

$$\Rightarrow 2x = 8$$

$$\Rightarrow x = 4$$

Let us assume that 'y' students are enrolled in K and L but not G. Then, the number of students enrolled only in K = y + 4

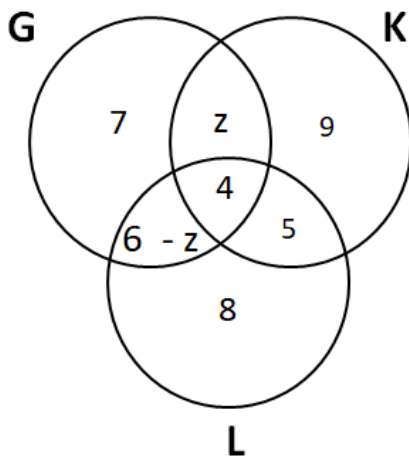
Let us assume that 'z' be the the number of students enrolled in G and K but not L. Then, the number of students enrolled G and L bot not K = 10 - 4 - z = 6 - z



It is given that a total of 39 students in the sports.

$$7 + z + 4 + 6 - z + 8 + y + y + 4 = 39$$

$$\Rightarrow y = 5$$



Number of students enrolled in G = 17

Number of students enrolled in K =  $9 + 4 + 5 + z = 18 + z$

Number of students enrolled in L =  $6 - z + 4 + 5 + 8 = 23 - z$

It is given that the maximum student enrollment is in L.

$$\Rightarrow 23 - z > 18 + z$$

$$\Rightarrow 2z < 5$$

$$\Rightarrow z < 2.5$$

Therefore, we can say that z can take three values =  $\{0, 1, 2\}$

Hence, the number of students enrolled in K =  $18 + z = \{18, 19, 20\}$

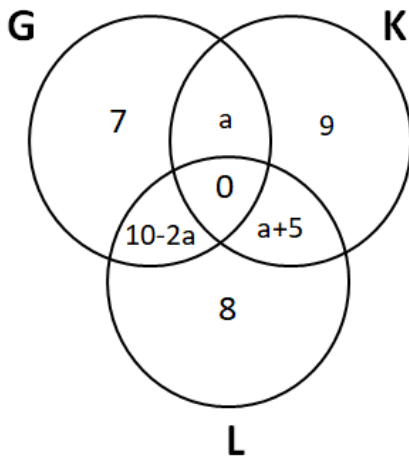
It is given that after withdrawal the number of students enrolled in K went down by one. This one student must have left sports K. Hence we can say that the remaining 3 students must have left either G or L.

Before withdraw there were a total of 24 students were enrolled in exactly 1 sports, 11 students were enrolled in exactly 2 courses and 4 students were enrolled in all three courses.

The students which were enrolled in all three sports, withdrew from one of the sports. Hence, we can say that now the number of students who were enrolled in exactly 2 courses =  $11 + 4 = 15$ .

It is given that the number of students enrolled in G was six less than the number of students enrolled in L. Let 'a' be the number of students who were enrolled in G and K but not L. Then, the number of students who were enrolled in L and K but not G =  $a + 5$

Consequently, we can say that the number of students enrolled in G and L but not K =  $15 - (2a + 5) = 10 - 2a$



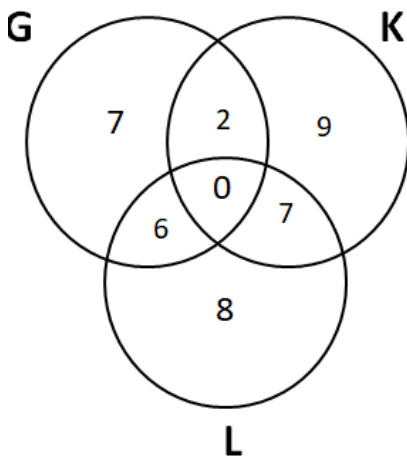
Number of students enrolled in this case =  $a + a+5 + 9 = 14 + 2a$ . We can see that ' $14+2a$ ' is an even number. It is given that the number of students enrolled in K went down by one. Therefore, we can say that the number of students enrolled in K earlier was an odd number.

Hence, the number of students enrolled in K =  $18 + z = \{18, 19, 20\}$

We can see that only '19' is an odd number. Hence, we can say that the number of students enrolled in K after withdrawal = 18

$$\Rightarrow 14 + 2a = 18$$

$$\Rightarrow a = 2$$



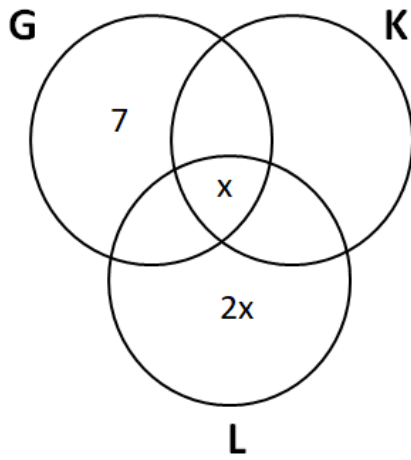
From the diagram we can see that the number of students enrolled in both G and K = 2.

[VIDEO SOLUTION](#)

20. A

Let 'x' be the number of students enrolled in all three sports. Then the number of students enrolled only in L =  $2x$

It is given that there are a total of 17 students enrolled in G. Also, ten students enrolled in G are also enrolled in at least one more sport. Hence, the number of students enrolled in only G =  $17 - 10 = 7$



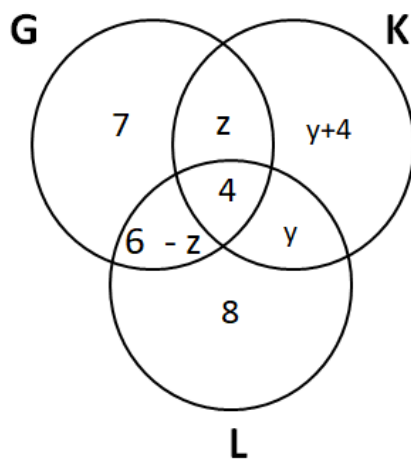
The number of students enrolled only in G is one less than the number of students enrolled only in L. Hence, the number of students enrolled only in L =  $7+1$

$$\Rightarrow 2x = 8$$

$$\Rightarrow x = 4$$

Let us assume that 'y' students are enrolled in K and L but not G. Then, the number of students enrolled only in K =  $y + 4$

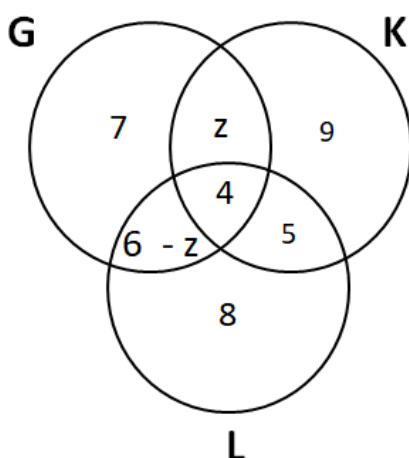
Let us assume that 'z' be the the number of students enrolled in G and K but not L. Then, the number of students enrolled G and L bot not K =  $10 - 4 - z = 6 - z$



It is given that a total of 39 students in the sports.

$$7 + z + 4 + 6 - z + 8 + y + y + 4 = 39$$

$$\Rightarrow y = 5$$



Number of students enrolled in G = 17

Number of students enrolled in K =  $9 + 4 + 5 + z = 18 + z$

Number of students enrolled in L =  $6 - z + 4 + 5 + 8 = 23 - z$

It is given that the maximum student enrollment is in L.

$$\Rightarrow 23 - z > 18 + z$$

$$\Rightarrow 2z < 5$$

$$\Rightarrow z < 2.5$$

Therefore, we can say that  $z$  can take three values =  $\{0, 1, 2\}$

Hence, the number of students enrolled in K =  $18 + z = \{18, 19, 20\}$

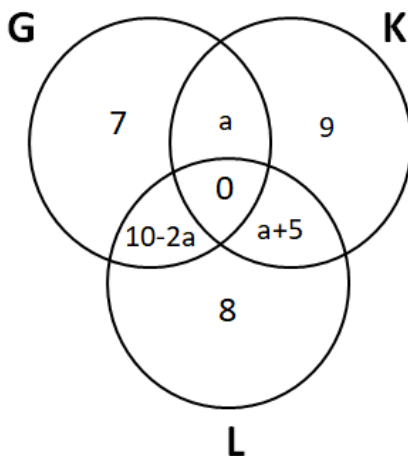
It is given that after withdrawal the number of students enrolled in K went down by one. This one student must have left sports K. Hence we can say that the remaining 3 students must have left either G or L.

Before withdraw there were a total of 24 students were enrolled in exactly 1 sports, 11 students were enrolled in exactly 2 courses and 4 students were enrolled in all three courses.

The students which were enrolled in all three sports, withdrew from one of the sports. Hence, we can say that now the number of students who were enrolled in exactly 2 courses =  $11 + 4 = 15$ .

It is given that the number of students enrolled in G was six less than the number of students enrolled in L. Let 'a' be the number of students who were enrolled in G and K but not L. Then, the number of students who were enrolled in L and K but not G =  $a + 5$

Consequently, we can say that the number of students enrolled in G and L but not K =  $15 - (2a + 5) = 10 - 2a$



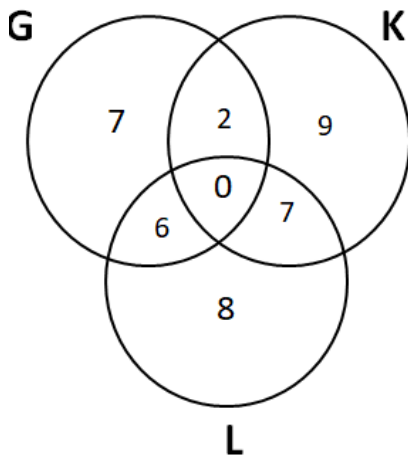
Number of students enrolled in this case =  $a + a + 5 + 9 = 14 + 2a$ . We can see that ' $14 + 2a$ ' is an even number. It is given that the number of students enrolled in K went down by one. Therefore, we can say that the number of students enrolled in K earlier was an odd number.

Hence, the number of students enrolled in K =  $18 + z = \{18, 19, 20\}$

We can see that only '19' is an odd number. Hence, we can say that the number of students enrolled in K after withdrawal = 18

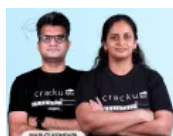
$$\Rightarrow 14 + 2a = 18$$

$$\Rightarrow a = 2$$

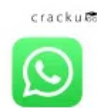


From the diagram we can see that the number of students enrolled in both G and L = 6. Hence, option A is the correct answer.

[VIDEO SOLUTION](#)



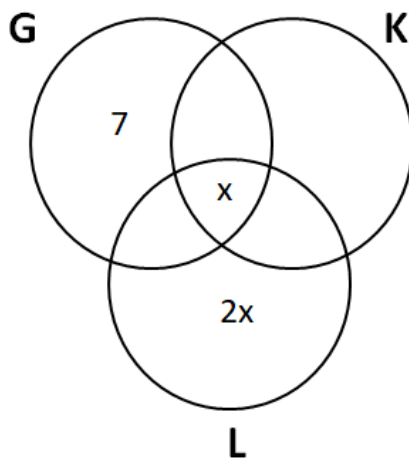
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21. C

Let 'x' be the number of students enrolled in all three sports. Then the number of students enrolled only in L =  $2x$

It is given that there are a total of 17 students enrolled in G. Also, ten students enrolled in G are also enrolled in at least one more sport. Hence, the number of students enrolled in only G =  $17 - 10 = 7$



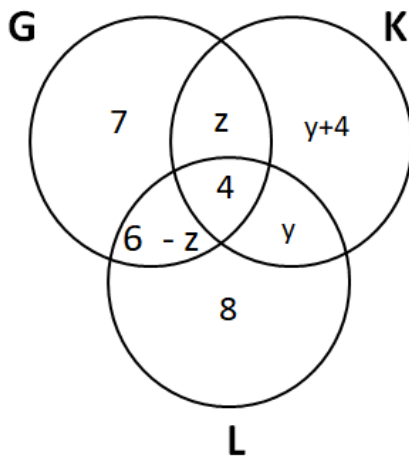
The number of students enrolled only in G is one less than the number of students enrolled only in L. Hence, the number of students enrolled only in L =  $7 + 1$

$$\Rightarrow 2x = 8$$

$$\Rightarrow x = 4$$

Let us assume that 'y' students are enrolled in K and L but not G. Then, the number of students enrolled only in K =  $y + 4$

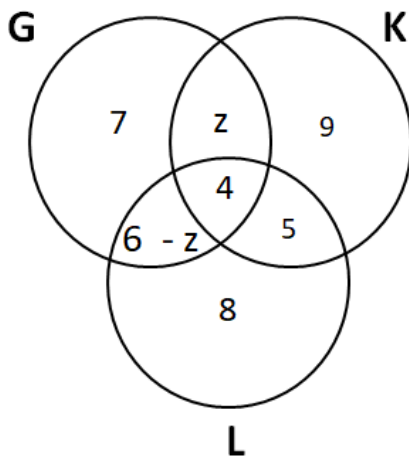
Let us assume that 'z' be the the number of students enrolled in G and K but not L. Then, the number of students enrolled G and L bot not K =  $10 - 4 - z = 6 - z$



It is given that a total of 39 students in the sports.

$$7 + z + 4 + 6 - z + 8 + y + y + 4 = 39$$

$$\Rightarrow y = 5$$



Number of students enrolled in G = 17

Number of students enrolled in K =  $9 + 4 + 5 + z = 18 + z$

Number of students enrolled in L =  $6 - z + 4 + 5 + 8 = 23 - z$

It is given that the maximum student enrollment is in L.

$$\Rightarrow 23 - z > 18 + z$$

$$\Rightarrow 2z < 5$$

$$\Rightarrow z < 2.5$$

Therefore, we can say that z can take three values = {0, 1, 2}

It is given that the numbers of students enrolled in K and L are in the ratio 19:22.

$$\frac{18 + z}{23 - z} = \frac{19}{22}$$

$z = 1$  which is a possible solution as well.

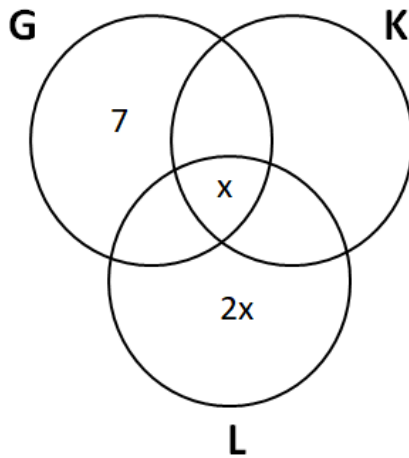
In this case the number of students enrolled in L =  $23 - z = 23 - 1 = 22$ . Hence, option C is the correct answer.

 VIDEO SOLUTION

22.4

Let 'x' be the number of students enrolled in all three sports. Then the number of students enrolled only in L =  $2x$

It is given that there are a total of 17 students enrolled in G. Also, ten students enrolled in G are also enrolled in at least one more sport. Hence, the number of students enrolled in only G =  $17 - 10 = 7$



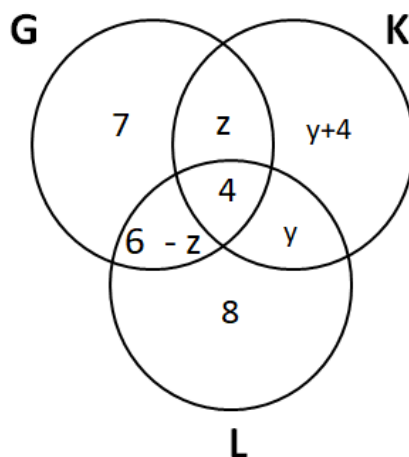
The number of students enrolled only in G is one less than the number of students enrolled only in L. Hence, the number of students enrolled only in L =  $7 + 1$

$$\Rightarrow 2x = 8$$

$$\Rightarrow x = 4$$

Let us assume that 'y' students are enrolled in K and L but not G. Then, the number of students enrolled only in K =  $y + 4$

Let us assume that 'z' be the the number of students enrolled in G and K but not L. Then, the number of students enrolled G and L bot not K =  $10 - 4 - z = 6 - z$

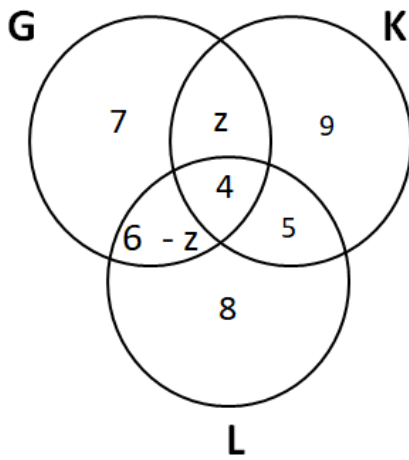


It is given that a total of 39 students in the sports.

$$7 + z + 4 + 6 - z + 8 + y + y + 4 = 39$$

$$\Rightarrow y = 5$$





Number of students enrolled in G = 17

Number of students enrolled in K =  $9 + 4 + 5 + z = 18 + z$

Number of students enrolled in L =  $6 - z + 4 + 5 + 8 = 23 - z$

It is given that the maximum student enrollment is in L.

$$\Rightarrow 23 - z > 18 + z$$

$$\Rightarrow 2z < 5$$

$$\Rightarrow z < 2.5$$

Therefore, we can say that z can take three values =  $\{0, 1, 2\}$

The number of students enrolled in both G and L but not in K =  $6 - z$ . This number will be minimum when 'z' is maximum.  $z_{\text{max}} = 2$

Therefore, the minimum number of students enrolled in both G and L but not in K =  $6 - 2 = 4$

 VIDEO SOLUTION

23. **A**

No. of girls = 15

Let the no. of boys be x

No. of singers = 6

No of boys who are singers = 4

Therefore, no of girls who are singers = 2

No of dancers = 10

No of boys who are dancers = 6

Therefore, no. of girls who are dancers = 4

No. of boys who are neither singers nor dancers =  $x - 10$

No. of girls who are neither singers nor dancers = 9

|                                    | Not interested | 1-day event | 2-day event | 2-day event |
|------------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (x)</b>                    |                |             |             |             |
| Singers (4)                        |                |             |             |             |
| Dancers (6)                        |                |             |             |             |
| Neither singers nor dancers (x-10) |                |             |             |             |
| <b>Girls (15)</b>                  |                |             |             |             |
| Singers (2)                        |                |             |             |             |
| Dancers (4)                        |                |             |             |             |
| Neither singers nor dancers (9)    |                |             |             |             |

Now we fill the above table,  
using statements 1 and 2, we get the following table

|                                    | Not interested | 1-day event | 2-day event | 3-day event |
|------------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (x)</b>                    | 0.2x           | 0.8x        | 0.6x        |             |
| Singers (4)                        |                |             |             |             |
| Dancers (6)                        |                |             |             |             |
| Neither singers nor dancers (x-10) |                |             |             |             |
| <b>Girls (15)</b>                  | 0              | 15          |             |             |
| Singers (2)                        | 0              | 2           |             |             |
| Dancers (4)                        | 0              | 4           |             |             |
| Neither singers nor dancers (9)    | 0              | 9           |             |             |

Let the number of girls who are interested in attending a 2-day event be a and the number of girls who are dancers and are interested in 2-day event be b.

Now using statements 3 and 4, we get

|                                    | Not interested | 1-day event | 2-day event | 3-day event |
|------------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (x)</b>                    | 0.2x           | 0.8x        | 0.6x        |             |
| Singers (4)                        | 0              | 4           | 4           | 4           |
| Dancers (6)                        |                |             | 0.18x-4     | 2           |
| Neither singers nor dancers (x-10) |                |             | 0.42x       |             |
| <b>Girls (15)</b>                  | 0              | 15          | a           | 0           |
| Singers (2)                        | 0              | 2           | 0.4a-b      | 0           |
| Dancers (4)                        | 0              | 4           | b           | 0           |
| Neither singers nor dancers (9)    | 0              | 9           | 0.6a        | 0           |

$$2 \leq 0.18x - 4 \leq 6$$

$$6 \leq 0.18x \leq 10$$

0.18x should be integer for which x should be a multiple of 50, and 0.18x lies between 6 and 10; therefore, the only possible value of x is 50.

|                                  | Not interested | 1-day event | 2-day event | 3-day event |
|----------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (50)</b>                 | 10             | 40          | 30          |             |
| Singers (4)                      | 0              | 4           | 4           | 4           |
| Dancers (6)                      |                |             | 5           | 2           |
| Neither singers nor dancers (40) |                |             | 21          |             |
| <b>Girls (15)</b>                | 0              | 15          | a           | 0           |
| Singers (2)                      | 0              | 2           | 0.4a-b      | 0           |
| Dancers (4)                      | 0              | 4           | b           | 0           |
| Neither singers nor dancers (9)  | 0              | 9           | 0.6a        | 0           |

From statement 5, we can say that,

$$4 + 0.4a - b = 5 + b + 1$$

$$\text{or, } 0.4a = 2 + 2b$$

$$\text{or, } a = 5(1+b)$$

a should be a multiple of 5 as b is a whole number. So possible values of a can be 5, 10 or 15. Now, as the maximum value of b can be 4 and the maximum value of 0.4a-b can be 2, so the only possible value of a satisfying the conditions above is 5. If a= 5 then b=1.

|                                  | Not interested | 1-day event | 2-day event | 3-day event |
|----------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (50)</b>                 | 10             | 40          | 30          |             |
| Singers (4)                      | 0              | 4           | 4           | 4           |
| Dancers (6)                      | 0/1            | 6/5         | 5           | 2           |
| Neither singers nor dancers (40) | 10/9           | 30/31       | 21          |             |
| <b>Girls (15)</b>                | 0              | 15          | 5           | 0           |
| Singers (2)                      | 0              | 2           | 2           | 0           |
| Dancers (4)                      | 0              | 4           | 0           | 0           |
| Neither singers nor dancers (9)  | 0              | 9           | 3           | 0           |

VIDEO SOLUTION

24.50

No. of girls = 15

Let the no. of boys be  $x$

No. of singers = 6

No of boys who are singers = 4

Therefore, no of girls who are singers = 2

No of dancers = 10

No of boys who are dancers = 6

Therefore, no. of girls who are dancers = 4

No. of boys who are neither singers nor dancers =  $x-10$

No. of girls who are neither singers nor dancers = 9

|                                    | Not interested | 1-day event | 2-day event | 2-day event |
|------------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (x)</b>                    |                |             |             |             |
| Singers (4)                        |                |             |             |             |
| Dancers (6)                        |                |             |             |             |
| Neither singers nor dancers (x-10) |                |             |             |             |
| <b>Girls (15)</b>                  |                |             |             |             |
| Singers (2)                        |                |             |             |             |
| Dancers (4)                        |                |             |             |             |
| Neither singers nor dancers (9)    |                |             |             |             |

Now we fill the above table,  
using statements 1 and 2, we get the following table

|                                    | Not interested | 1-day event | 2-day event | 3-day event |
|------------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (x)</b>                    | 0.2x           | 0.8x        | 0.6x        |             |
| Singers (4)                        |                |             |             |             |
| Dancers (6)                        |                |             |             |             |
| Neither singers nor dancers (x-10) |                |             |             |             |
| <b>Girls (15)</b>                  | 0              | 15          |             |             |
| Singers (2)                        | 0              | 2           |             |             |
| Dancers (4)                        | 0              | 4           |             |             |
| Neither singers nor dancers (9)    | 0              | 9           |             |             |

Let the number of girls who are interested in attending a 2-day event be a and the number of girls who are dancers and are interested in 2-day event be b.

Now using statements 3 and 4, we get



|                                  | Not interested | 1-day event | 2-day event | 3-day event |
|----------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (50)</b>                 | 10             | 40          | 30          |             |
| Singers (4)                      | 0              | 4           | 4           | 4           |
| Dancers (6)                      | 0/1            | 6/5         | 5           | 2           |
| Neither singers nor dancers (40) | 10/9           | 30/31       | 21          |             |
| <b>Girls (15)</b>                | 0              | 15          | 5           | 0           |
| Singers (2)                      | 0              | 2           | 2           | 0           |
| Dancers (4)                      | 0              | 4           | 0           | 0           |
| Neither singers nor dancers (9)  | 0              | 9           | 3           | 0           |

 VIDEO SOLUTION

25. C

No. of girls = 15

Let the no. of boys be  $x$

No. of singers = 6

No of boys who are singers = 4

Therefore, no of girls who are singers = 2

No of dancers = 10

No of boys who are dancers = 6

Therefore, no. of girls who are dancers = 4

No. of boys who are neither singers nor dancers =  $x-10$

No. of girls who are neither singers nor dancers = 9

|                                    | Not interested | 1-day event | 2-day event | 2-day event |
|------------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (x)</b>                    |                |             |             |             |
| Singers (4)                        |                |             |             |             |
| Dancers (6)                        |                |             |             |             |
| Neither singers nor dancers (x-10) |                |             |             |             |
| <b>Girls (15)</b>                  |                |             |             |             |
| Singers (2)                        |                |             |             |             |
| Dancers (4)                        |                |             |             |             |
| Neither singers nor dancers (9)    |                |             |             |             |

Now we fill the above table,  
using statements 1 and 2, we get the following table

|                                    | Not interested | 1-day event | 2-day event | 3-day event |
|------------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (x)</b>                    | 0.2x           | 0.8x        | 0.6x        |             |
| Singers (4)                        |                |             |             |             |
| Dancers (6)                        |                |             |             |             |
| Neither singers nor dancers (x-10) |                |             |             |             |
| <b>Girls (15)</b>                  | 0              | 15          |             |             |
| Singers (2)                        | 0              | 2           |             |             |
| Dancers (4)                        | 0              | 4           |             |             |
| Neither singers nor dancers (9)    | 0              | 9           |             |             |

Let the number of girls who are interested in attending a 2-day event be a and the number of girls who are dancers and are interested in 2-day event be b.

Now using statements 3 and 4, we get



|                                    | Not interested | 1-day event | 2-day event | 3-day event |
|------------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (x)</b>                    | 0.2x           | 0.8x        | 0.6x        |             |
| Singers (4)                        | 0              | 4           | 4           | 4           |
| Dancers (6)                        |                |             | 0.18x-4     | 2           |
| Neither singers nor dancers (x-10) |                |             | 0.42x       |             |
| <b>Girls (15)</b>                  | 0              | 15          | a           | 0           |
| Singers (2)                        | 0              | 2           | 0.4a-b      | 0           |
| Dancers (4)                        | 0              | 4           | b           | 0           |
| Neither singers nor dancers (9)    | 0              | 9           | 0.6a        | 0           |

$$2 \leq 0.18x - 4 \leq 6$$

$$6 \leq 0.18x \leq 10$$

0.18x should be integer for which x should be a multiple of 50, and 0.18x lies between 6 and 10; therefore, the only possible value of x is 50.

|                                  | Not interested | 1-day event | 2-day event | 3-day event |
|----------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (50)</b>                 | 10             | 40          | 30          |             |
| Singers (4)                      | 0              | 4           | 4           | 4           |
| Dancers (6)                      |                |             | 5           | 2           |
| Neither singers nor dancers (40) |                |             | 21          |             |
| <b>Girls (15)</b>                | 0              | 15          | a           | 0           |
| Singers (2)                      | 0              | 2           | 0.4a-b      | 0           |
| Dancers (4)                      | 0              | 4           | b           | 0           |
| Neither singers nor dancers (9)  | 0              | 9           | 0.6a        | 0           |

From statement 5, we can say that,

$$4 + 0.4a - b = 5 + b + 1$$

$$\text{or, } 0.4a = 2 + 2b$$

$$\text{or, } a = 5(1+b)$$

a should be a multiple of 5 as b is a whole number. So possible values of a can be 5, 10 or 15. Now, as the maximum value of b can be 4 and the maximum value of 0.4a-b can be 2, so the only possible value of a satisfying the conditions above is 5. If a= 5 then b=1.

|                                  | Not interested | 1-day event | 2-day event | 3-day event |
|----------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (50)</b>                 | 10             | 40          | 30          |             |
| Singers (4)                      | 0              | 4           | 4           | 4           |
| Dancers (6)                      | 0/1            | 6/5         | 5           | 2           |
| Neither singers nor dancers (40) | 10/9           | 30/31       | 21          |             |
| <b>Girls (15)</b>                | 0              | 15          | 5           | 0           |
| Singers (2)                      | 0              | 2           | 2           | 0           |
| Dancers (4)                      | 0              | 4           | 0           | 0           |
| Neither singers nor dancers (9)  | 0              | 9           | 3           | 0           |

VIDEO SOLUTION

**CAT Mock Test**

Mocks designed exactly like the actual CAT

26. C

No. of girls = 15

Let the no. of boys be  $x$

No. of singers = 6

No of boys who are singers = 4

Therefore, no of girls who are singers = 2

No of dancers = 10

No of boys who are dancers = 6

Therefore, no. of girls who are dancers = 4

No. of boys who are neither singers nor dancers =  $x-10$

No. of girls who are neither singers nor dancers = 9

|                                    | Not interested | 1-day event | 2-day event | 2-day event |
|------------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (x)</b>                    |                |             |             |             |
| Singers (4)                        |                |             |             |             |
| Dancers (6)                        |                |             |             |             |
| Neither singers nor dancers (x-10) |                |             |             |             |
| <b>Girls (15)</b>                  |                |             |             |             |
| Singers (2)                        |                |             |             |             |
| Dancers (4)                        |                |             |             |             |
| Neither singers nor dancers (9)    |                |             |             |             |

Now we fill the above table,  
using statements 1 and 2, we get the following table

|                                    | Not interested | 1-day event | 2-day event | 3-day event |
|------------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (x)</b>                    | 0.2x           | 0.8x        | 0.6x        |             |
| Singers (4)                        |                |             |             |             |
| Dancers (6)                        |                |             |             |             |
| Neither singers nor dancers (x-10) |                |             |             |             |
| <b>Girls (15)</b>                  | 0              | 15          |             |             |
| Singers (2)                        | 0              | 2           |             |             |
| Dancers (4)                        | 0              | 4           |             |             |
| Neither singers nor dancers (9)    | 0              | 9           |             |             |

Let the number of girls who are interested in attending a 2-day event be  $a$  and the number of girls who are dancers and are interested in 2-day event be  $b$ .

Now using statements 3 and 4, we get

|                                    | Not interested | 1-day event | 2-day event | 3-day event |
|------------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (x)</b>                    | 0.2x           | 0.8x        | 0.6x        |             |
| Singers (4)                        | 0              | 4           | 4           | 4           |
| Dancers (6)                        |                |             | $0.18x - 4$ | 2           |
| Neither singers nor dancers (x-10) |                |             | 0.42x       |             |
| <b>Girls (15)</b>                  | 0              | 15          | a           | 0           |
| Singers (2)                        | 0              | 2           | $0.4a - b$  | 0           |
| Dancers (4)                        | 0              | 4           | b           | 0           |
| Neither singers nor dancers (9)    | 0              | 9           | $0.6a$      | 0           |

$$2 \leq 0.18x - 4 \leq 6$$

$$6 \leq 0.18x \leq 10$$

0.18x should be integer for which x should be a multiple of 50, and 0.18x lies between 6 and 10; therefore, the only possible value of x is 50.

|                                  | Not interested | 1-day event | 2-day event | 3-day event |
|----------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (50)</b>                 | 10             | 40          | 30          |             |
| Singers (4)                      | 0              | 4           | 4           | 4           |
| Dancers (6)                      |                |             | 5           | 2           |
| Neither singers nor dancers (40) |                |             | 21          |             |
| <b>Girls (15)</b>                | 0              | 15          | a           | 0           |
| Singers (2)                      | 0              | 2           | $0.4a - b$  | 0           |
| Dancers (4)                      | 0              | 4           | b           | 0           |
| Neither singers nor dancers (9)  | 0              | 9           | $0.6a$      | 0           |

From statement 5, we can say that,

$$4 + 0.4a - b = 5 + b + 1$$

$$\text{or, } 0.4a = 2 + 2b$$

$$\text{or, } a = 5(1 + b)$$

a should be a multiple of 5 as b is a whole number. So possible values of a can be 5, 10 or 15. Now, as the maximum value of b can be 4 and the maximum value of  $0.4a - b$  can be 2, so the only possible value of a satisfying the conditions above is 5. If  $a = 5$  then  $b = 1$ .

|                                  | Not interested | 1-day event | 2-day event | 3-day event |
|----------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (50)</b>                 | 10             | 40          | 30          |             |
| Singers (4)                      | 0              | 4           | 4           | 4           |
| Dancers (6)                      | 0/1            | 6/5         | 5           | 2           |
| Neither singers nor dancers (40) | 10/9           | 30/31       | 21          |             |
| <b>Girls (15)</b>                | 0              | 15          | 5           | 0           |
| Singers (2)                      | 0              | 2           | 2           | 0           |
| Dancers (4)                      | 0              | 4           | 0           | 0           |
| Neither singers nor dancers (9)  | 0              | 9           | 3           | 0           |

VIDEO SOLUTION

27. B

No. of girls = 15

Let the no. of boys be  $x$

No. of singers = 6

No of boys who are singers = 4

Therefore, no of girls who are singers = 2

No of dancers = 10

No of boys who are dancers = 6

Therefore, no. of girls who are dancers = 4

No. of boys who are neither singers nor dancers =  $x-10$

No. of girls who are neither singers nor dancers = 9

|                                    | Not interested | 1-day event | 2-day event | 2-day event |
|------------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (x)</b>                    |                |             |             |             |
| Singers (4)                        |                |             |             |             |
| Dancers (6)                        |                |             |             |             |
| Neither singers nor dancers (x-10) |                |             |             |             |
| <b>Girls (15)</b>                  |                |             |             |             |
| Singers (2)                        |                |             |             |             |
| Dancers (4)                        |                |             |             |             |
| Neither singers nor dancers (9)    |                |             |             |             |

Now we fill the above table,  
using statements 1 and 2, we get the following table

|                                    | Not interested | 1-day event | 2-day event | 3-day event |
|------------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (x)</b>                    | 0.2x           | 0.8x        | 0.6x        |             |
| Singers (4)                        |                |             |             |             |
| Dancers (6)                        |                |             |             |             |
| Neither singers nor dancers (x-10) |                |             |             |             |
| <b>Girls (15)</b>                  | 0              | 15          |             |             |
| Singers (2)                        | 0              | 2           |             |             |
| Dancers (4)                        | 0              | 4           |             |             |
| Neither singers nor dancers (9)    | 0              | 9           |             |             |

Let the number of girls who are interested in attending a 2-day event be a and the number of girls who are dancers and are interested in 2-day event be b.

Now using statements 3 and 4, we get

|                                    | Not interested | 1-day event | 2-day event | 3-day event |
|------------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (x)</b>                    | 0.2x           | 0.8x        | 0.6x        |             |
| Singers (4)                        | 0              | 4           | 4           | 4           |
| Dancers (6)                        |                |             | $0.18x - 4$ | 2           |
| Neither singers nor dancers (x-10) |                |             | 0.42x       |             |
| <b>Girls (15)</b>                  | 0              | 15          | a           | 0           |
| Singers (2)                        | 0              | 2           | $0.4a - b$  | 0           |
| Dancers (4)                        | 0              | 4           | b           | 0           |
| Neither singers nor dancers (9)    | 0              | 9           | $0.6a$      | 0           |

$$2 \leq 0.18x - 4 \leq 6$$

$$6 \leq 0.18x \leq 10$$

0.18x should be integer for which x should be a multiple of 50, and 0.18x lies between 6 and 10; therefore, the only possible value of x is 50.

|                                  | Not interested | 1-day event | 2-day event | 3-day event |
|----------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (50)</b>                 | 10             | 40          | 30          |             |
| Singers (4)                      | 0              | 4           | 4           | 4           |
| Dancers (6)                      |                |             | 5           | 2           |
| Neither singers nor dancers (40) |                |             | 21          |             |
| <b>Girls (15)</b>                | 0              | 15          | a           | 0           |
| Singers (2)                      | 0              | 2           | $0.4a - b$  | 0           |
| Dancers (4)                      | 0              | 4           | b           | 0           |
| Neither singers nor dancers (9)  | 0              | 9           | $0.6a$      | 0           |

From statement 5, we can say that,

$$4 + 0.4a - b = 5 + b + 1$$

$$\text{or, } 0.4a = 2 + 2b$$

$$\text{or, } a = 5(1 + b)$$

a should be a multiple of 5 as b is a whole number. So possible values of a can be 5, 10 or 15. Now, as the maximum value of b can be 4 and the maximum value of  $0.4a - b$  can be 2, so the only possible value of a satisfying the conditions above is 5. If  $a = 5$  then  $b = 1$ .

|                                  | Not interested | 1-day event | 2-day event | 3-day event |
|----------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (50)</b>                 | 10             | 40          | 30          |             |
| Singers (4)                      | 0              | 4           | 4           | 4           |
| Dancers (6)                      | 0/1            | 6/5         | 5           | 2           |
| Neither singers nor dancers (40) | 10/9           | 30/31       | 21          |             |
| <b>Girls (15)</b>                | 0              | 15          | 5           | 0           |
| Singers (2)                      | 0              | 2           | 2           | 0           |
| Dancers (4)                      | 0              | 4           | 0           | 0           |
| Neither singers nor dancers (9)  | 0              | 9           | 3           | 0           |

Total students interested in 2-day event = 30+5 = 35

Total students = 15 + 50 = 65

$$\text{Fraction} = \frac{35}{65} = \frac{7}{13}$$

VIDEO SOLUTION

28. A

Total number of students surveyed= 500

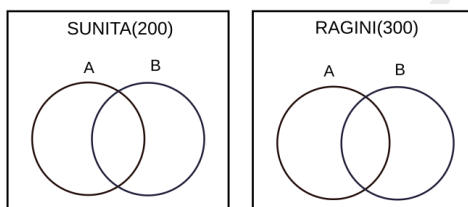
Every student prefers one of the two candidates. Ragini(R) and Sunita(S).

Thus, R+S=500.

According to statement 2, "Among the 200 students who preferred Sunita as student union president, 80% supported proposal A."

The number of students who support Sunita(S)=200

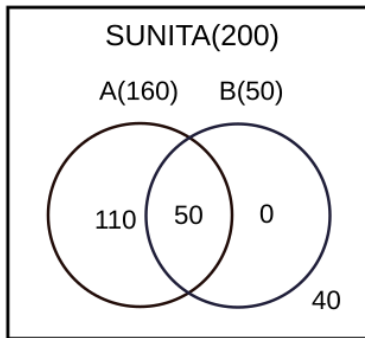
The number of students who supported Ragini(R)=300



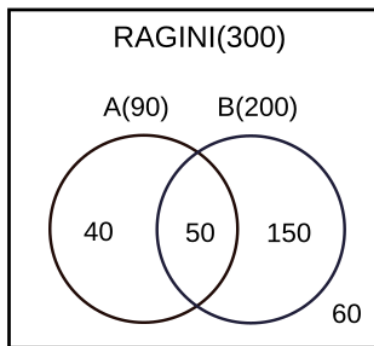
According to statements 2 and 3, 160 students who supported Sunita also supported the proposal A & 90 students who supported Ragini also supported proposal A.

According to statements 4 and 6, we can make the following Venn diagram for Sunita.





According to statement 5 and 7, we can make the following Venn diagram.



From the diagram, we can understand that option A is correct.

[VIDEO SOLUTION](#)

29. D

Total number of students surveyed = 500

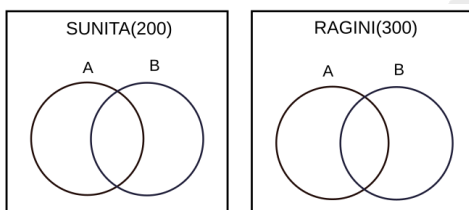
Every student prefers one of the two candidates. Ragini(R) and Sunita(S).

Thus,  $R + S = 500$ .

According to statement 2, "Among the 200 students who preferred Sunita as student union president, 80% supported proposal A."

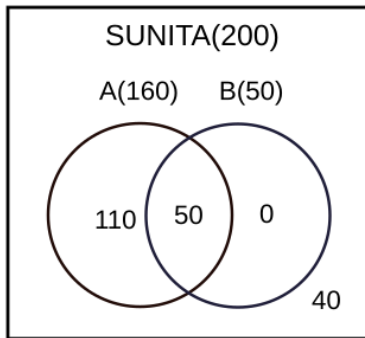
The number of students who support Sunita(S) = 200

The number of students who supported Ragini(R) = 300

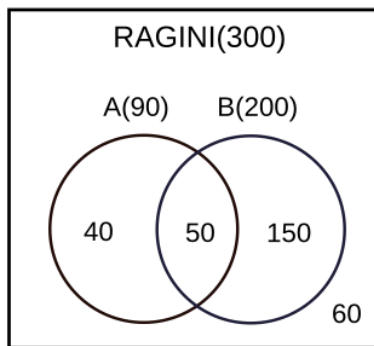


According to statements 2 and 3, 160 students who supported Sunita also supported the proposal A & 90 students who supported Ragini also supported proposal A.

According to statements 4 and 6, we can make the following Venn diagram for Sunita.



According to statement 5 and 7, we can make the following Venn diagram.



According to the Venn diagram, the students surveyed who supported both proposals A and B preferred Sunita as student union president  $\frac{50}{50+50} \% = 50\%$

[VIDEO SOLUTION](#)

**30.84**

Total number of students surveyed = 500

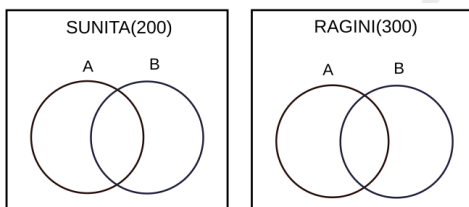
Every student prefers one of the two candidates. Ragini(R) and Sunita(S).

Thus,  $R+S=500$ .

According to statement 2, "Among the 200 students who preferred Sunita as student union president, 80% supported proposal A."

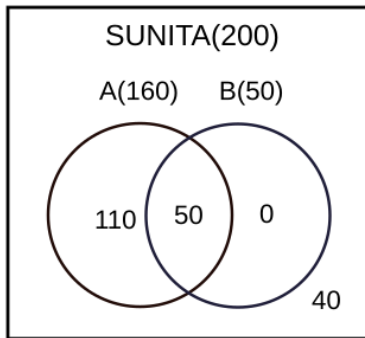
The number of students who support Sunita(S)=200

The number of students who supported Ragini(R)=300

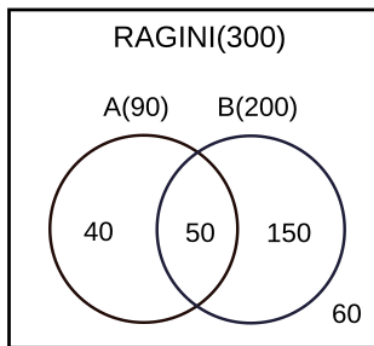


According to statements 2 and 3, 160 students who supported Sunita also supported the proposal A & 90 students who supported Ragini also supported proposal A.

According to statements 4 and 6, we can make the following Venn diagram for Sunita.



According to statement 5 and 7, we can make the following Venn diagram.



The percentage of the students surveyed who did not support proposal A preferred Ragini as student union president =  $210/250 = 84\%$

Answer 84

[VIDEO SOLUTION](#)



## CAT Online Coaching

your road to IIM stars here

31.64

Total number of students surveyed = 500

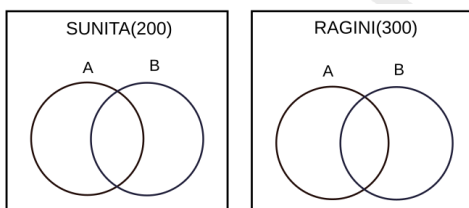
Every student prefers one of the two candidates. Ragini(R) and Sunita(S).

Thus,  $R + S = 500$ .

According to statement 2, "Among the 200 students who preferred Sunita as student union president, 80% supported proposal A."

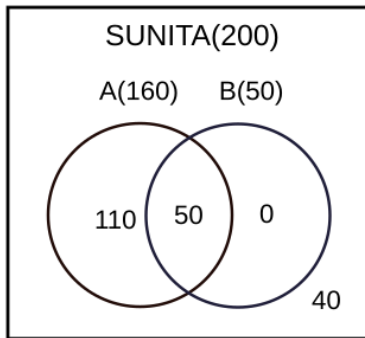
The number of students who support Sunita(S) = 200

The number of students who supported Ragini(R) = 300

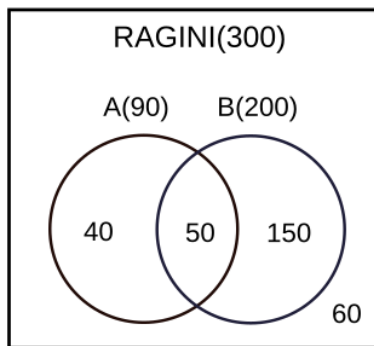


According to statements 2 and 3, 160 students who supported Sunita also supported the proposal A & 90 students who supported Ragini also supported proposal A.

According to statements 4 and 6, we can make the following Venn diagram for Sunita.



According to statement 5 and 7, we can make the following Venn diagram.



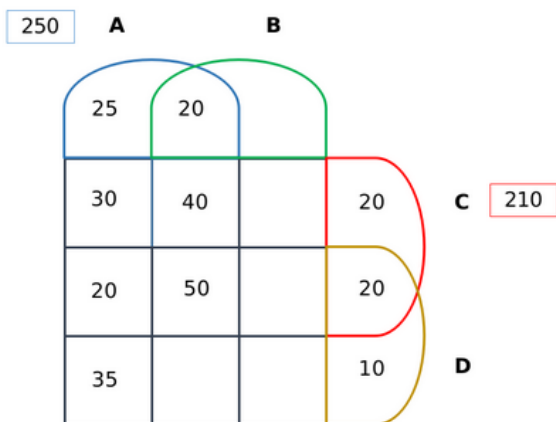
The number of students who preferred Sunita and the proposal A=160

$$=160/250= 64\%$$

[VIDEO SOLUTION](#)

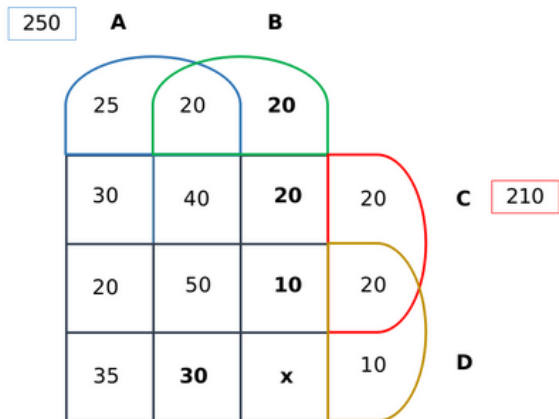
### 32. 325

Of the 1000 subjects, only 500 have been considered for the treatment. This constitutes our sample set. Thus the four drugs- A, B, C and D have been administered to this set of 500 individuals, while the rest 500 have been given the placebo. Based on the given information, we can then draw the following 4-set Venn diagram:

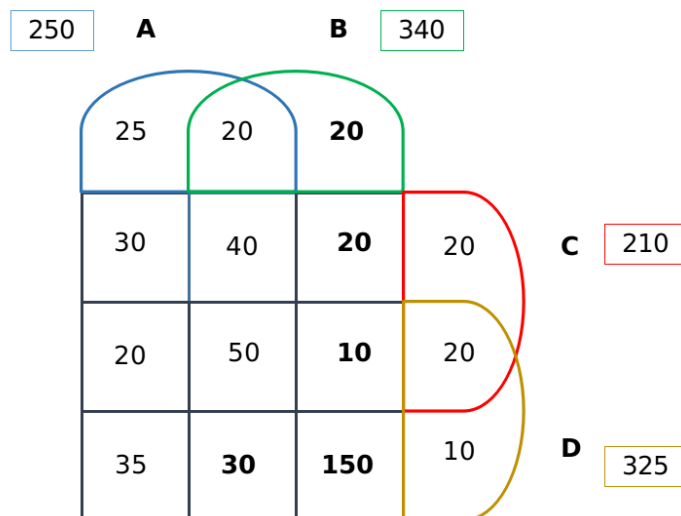


We can solve for the number of patients who were administered the drugs A, B and D excluding C by putting in the values for set A. The required value =  $250 - (25+20+30+40+20+50+35) = 30$ . Based on condition (c), we know that 100 patients were treated with exactly three types of medicines. Thus, we can fill the slot for the number of patients who were administered only B, C and D excluding A by  $100 - (40+20+30) = 10$ .

Similarly, based on condition (f), we know that the candidates who were administered only drug B are  $75 - (25+20+10) = 20$ . Post this, we can easily calculate the number of people administered with only drugs B and C by  $210 - (30+20+40+50+10+20+20) = 20$ . We can fill in the above values to obtain the following diagram:



The sum of all the values should add up to 500. On solving for 'x' [which represents the number of people who were administered drugs B and D only], we obtain  $x = 150$ . The final representation would appear as follows:



The number of patients who were treated with medicine type D was **325**.

[VIDEO SOLUTION](#)

**33. 340**

Of the 1000 subjects, only 500 have been considered for the treatment. This constitutes our sample set. Thus the four drugs- A, B, C and D have been administered to this set of 500 individuals, while the rest 500 have been given the placebo. Based on the given information, we can then draw the following 4-set Venn diagram:

|     |          |          |  |          |
|-----|----------|----------|--|----------|
| 250 | <b>A</b> | <b>B</b> |  |          |
|     | 25       | 20       |  |          |
|     | 30       | 40       |  | 20       |
|     | 20       | 50       |  | 20       |
|     | 35       |          |  | 10       |
|     |          |          |  | <b>D</b> |

**C** 210

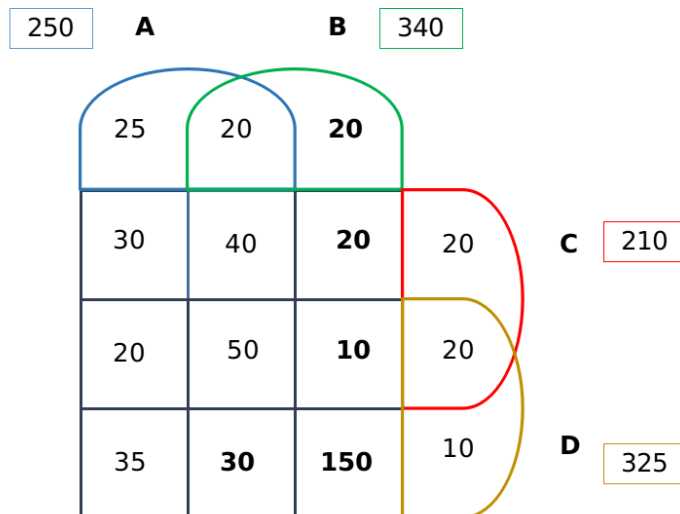
We can solve for the number of patients who were administered the drugs A, B and D excluding C by putting in the values for set A. The required value =  $250 - (25+20+30+40+20+50+35) = 30$ . Based on condition (c), we know that 100 patients were treated with exactly three types of medicines. Thus, we can fill the slot for the number of patients who were administered only B, C and D excluding A by  $100 - (40+20+30) = 10$ .

Similarly, based on condition (f), we know that the candidates who were administered only drug B are  $75 - (25+20+10) = 20$ . Post this, we can easily calculate the number of people administered with only drugs B and C by  $210 - (30+20+40+50+10+20+20) = 20$ . We can fill in the above values to obtain the following diagram:

|     |          |          |    |          |
|-----|----------|----------|----|----------|
| 250 | <b>A</b> | <b>B</b> |    |          |
|     | 25       | 20       | 20 |          |
|     | 30       | 40       | 20 | 20       |
|     | 20       | 50       | 10 | 20       |
|     | 35       | 30       | x  | 10       |
|     |          |          |    | <b>D</b> |

**C** 210

The sum of all the values should add up to 500. On solving for 'x' [which represents the number of people who were administered drugs B and D only], we obtain  $x = 150$ . The final representation would appear as follows:

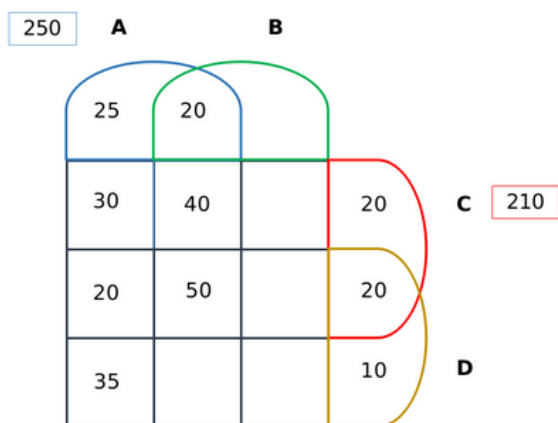


Based on the above, the number of patients who were treated with medicine type B is equal to **340**.

[VIDEO SOLUTION](#)

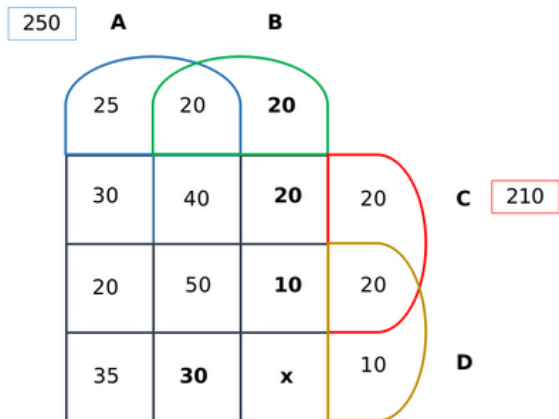
### 34.10

Of the 1000 subjects, only 500 have been considered for the treatment. This constitutes our sample set. Thus the four drugs- A, B, C and D have been administered to this set of 500 individuals, while the rest 500 have been given the placebo. Based on the given information, we can then draw the following 4-set Venn diagram:

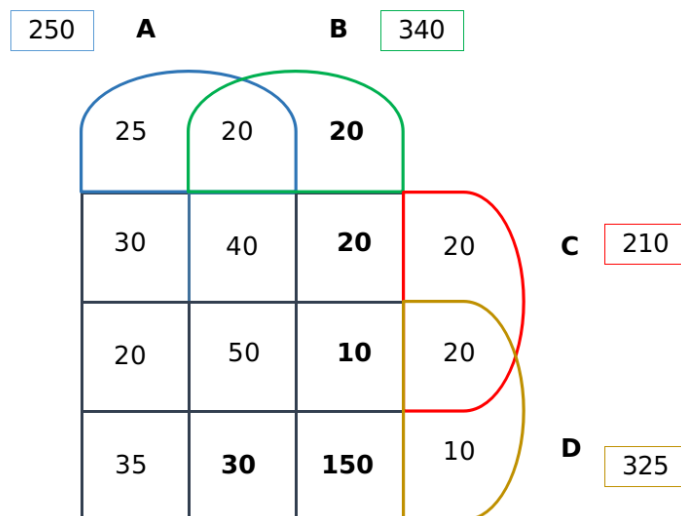


We can solve for the number of patients who were administered the drugs A, B and D excluding C by putting in the values for set A. The required value =  $250 - (25 + 20 + 30 + 40 + 20 + 50 + 35) = 30$ . Based on condition (c), we know that 100 patients were treated with exactly three types of medicines. Thus, we can fill the slot for the number of patients who were administered only B, C and D excluding A by  $100 - (40 + 20 + 30) = 10$ .

Similarly, based on condition (f), we know that the candidates who were administered only drug B are  $75 - (25 + 20 + 10) = 20$ . Post this, we can easily calculate the number of people administered with only drugs B and C by  $210 - (30 + 20 + 40 + 50 + 10 + 20 + 20) = 20$ . We can fill in the above values to obtain the following diagram:



The sum of all the values should add up to 500. On solving for 'x' [which represents the number of people who were administered drugs B and D only], we obtain  $x = 150$ . The final representation would appear as follows:



The number of patients who were treated with medicine types B, C and D, but not type A was: **10**.

[VIDEO SOLUTION](#)

35. **150**

Of the 1000 subjects, only 500 have been considered for the treatment. This constitutes our sample set. Thus the four drugs- A, B, C and D have been administered to this set of 500 individuals, while the rest 500 have been given the placebo. Based on the given information, we can then draw the following 4-set Venn diagram:



|     |          |          |  |          |
|-----|----------|----------|--|----------|
| 250 | <b>A</b> | <b>B</b> |  |          |
|     | 25       | 20       |  |          |
|     | 30       | 40       |  | 20       |
|     | 20       | 50       |  | 20       |
|     | 35       |          |  | 10       |
|     |          |          |  | <b>D</b> |

**C** 210

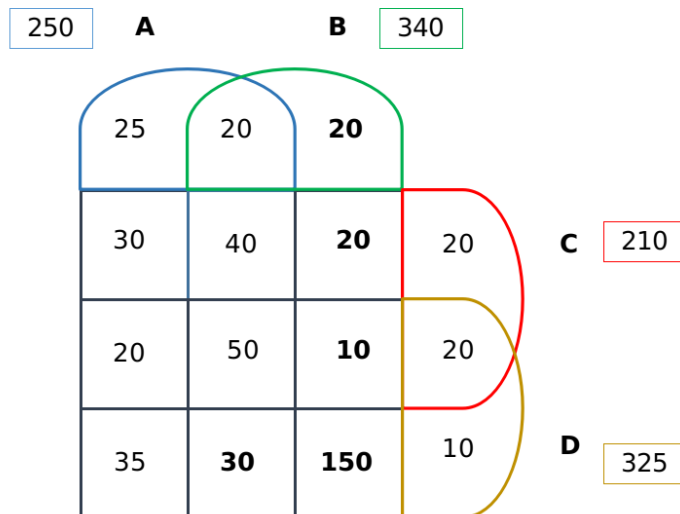
We can solve for the number of patients who were administered the drugs A, B and D excluding C by putting in the values for set A. The required value =  $250 - (25+20+30+40+20+50+35) = 30$ . Based on condition (c), we know that 100 patients were treated with exactly three types of medicines. Thus, we can fill the slot for the number of patients who were administered only B, C and D excluding A by  $100 - (40+20+30) = 10$ .

Similarly, based on condition (f), we know that the candidates who were administered only drug B are  $75 - (25+20+10) = 20$ . Post this, we can easily calculate the number of people administered with only drugs B and C by  $210 - (30+20+40+50+10+20+20) = 20$ . We can fill in the above values to obtain the following diagram:

|     |          |          |    |          |
|-----|----------|----------|----|----------|
| 250 | <b>A</b> | <b>B</b> |    |          |
|     | 25       | 20       | 20 |          |
|     | 30       | 40       | 20 | 20       |
|     | 20       | 50       | 10 | 20       |
|     | 35       | 30       | x  | 10       |
|     |          |          |    | <b>D</b> |

**C** 210

The sum of all the values should add up to 500. On solving for 'x' [which represents the number of people who were administered drugs B and D only], we obtain  $x = 150$ . The final representation would appear as follows:



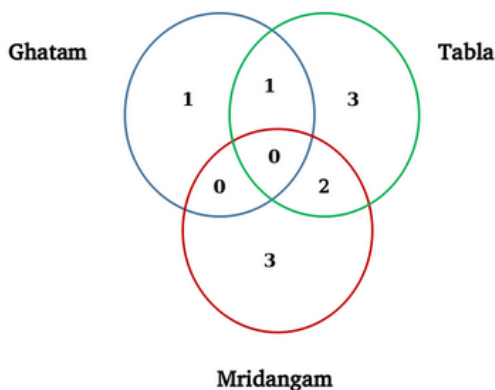
The number of people who were administered drugs B and D only were **150**.

[VIDEO SOLUTION](#)



36. D

Based on the given information, we can form the following Venn-diagram for ease of understanding:



The conditions help us to further bifurcate the individuals based on their expertise.

Mridangam: A and B (condition 1); one out of F and G (condition 3)

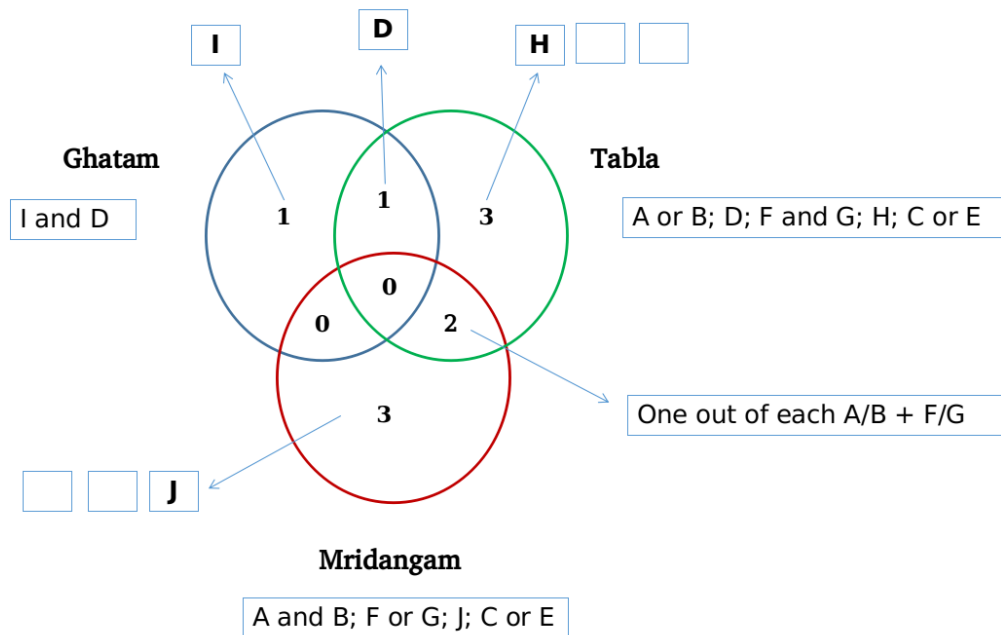
Tabla: one out of A and B (condition 1); F and G (condition 3); D (condition 2)

Ghatam: D (condition 2)

Based on condition 4, we infer that 'I' and 'J' are either experts in Ghatam or Mridangam. However, condition 5 adds that 'I' is not an expert in Mridangam. This helps us definitively zero-in on 'I' as an expert in Ghatam. Since 'I' is a Ghatam expert, J is an expert in Mridangam and 'H' is an expert in Tabla [based on conditions 4 and 5]. Thus, we can depict our understanding so far as follows:



Based on condition 4, we infer that 'I' and 'J' are either experts in Ghatam or Mridangam. However, condition 5 adds that 'I' is not an expert in Mridangam. This helps us definitively zero-in on 'I' as an expert in Ghatam. Since 'I' is a Ghatam expert, J is an expert in Mridangam and 'H' is an expert in Tabla [based on conditions 4 and 5]. Thus, we can depict our understanding so far as follows:



**Mridangam** [total: 5] - A and B (condition 1); one out of F and G (condition 3); J (condition 4 and 5); one out of C and E {remaining experts}

**Tabla** [total: 6] - one out of A and B (condition 1); F and G (condition 3); D (condition 2); H (condition 4 and 5); one out of C and E {remaining experts}

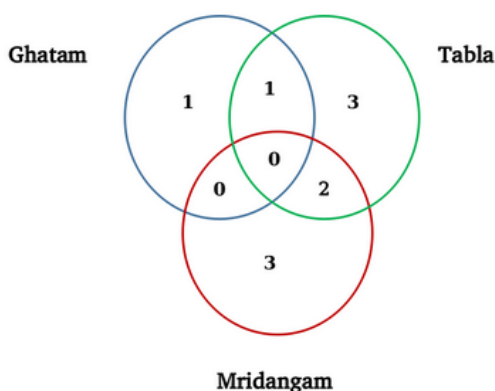
**Ghatam** [total: 2] - D (condition 2); I (condition 4 and 5)

We observe that the pair **C and E** cannot have any musician who is an expert in both tabla and mridangam but not in ghatam. Hence, Option B is the correct answer.

VIDEO SOLUTION

38. A

Based on the given information, we can form the following Venn-diagram for ease of understanding:



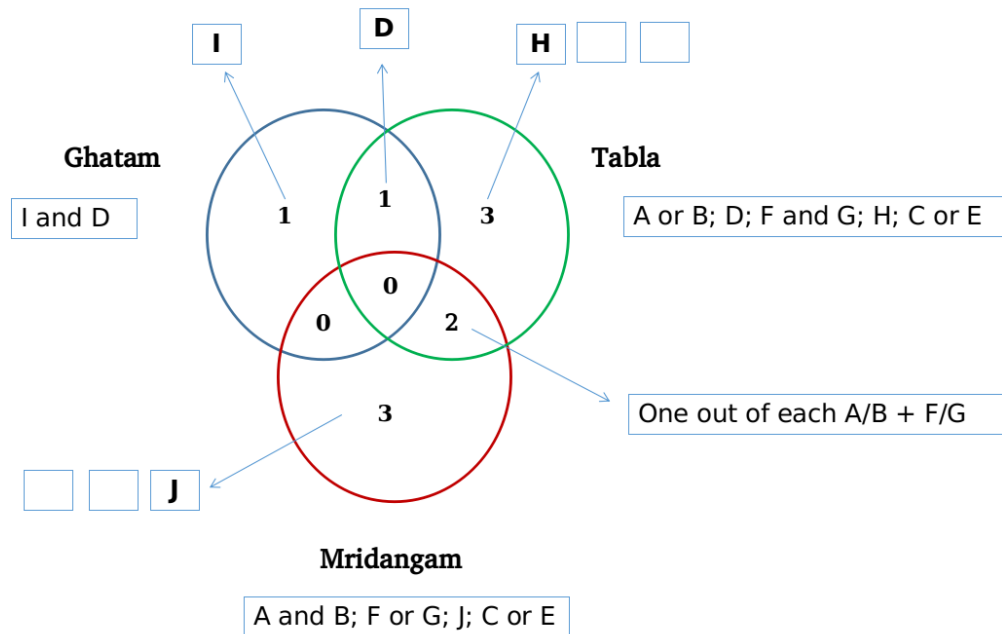
The conditions help us to further bifurcate the individuals based on their expertise.

Mridangam: A and B (condition 1); one out of F and G (condition 3)

Tabla: one out of A and B (condition 1); F and G (condition 3); D (condition 2)

Ghatam: D (condition 2)

Based on condition 4, we infer that 'I' and 'J' are either experts in Ghatam or Mridangam. However, condition 5 adds that 'I' is not an expert in Mridangam. This helps us definitively zero-in on 'I' as an expert in Ghatam. Since 'I' is a Ghatam expert, J is an expert in Mridangam and 'H' is an expert in Tabla [based on conditions 4 and 5]. Thus, we can depict our understanding so far as follows:



**Mridangam** [total: 5] - A and B (condition 1); one out of F and G (condition 3); J (condition 4 and 5); one out of C and E {remaining experts}

**Tabla** [total: 6] - one out of A and B (condition 1); F and G (condition 3); D (condition 2); H (condition 4 and 5); one out of C and E {remaining experts}

**Ghatam** [total: 2] - D (condition 2); I (condition 4 and 5)

If C is an expert in Mridangam, E has to be an expert in Tabla. Additionally, if F is not an expert in Mridangam, he has to be an expert only in Tabla while G will be an expert in both Tabla and Mridangam.

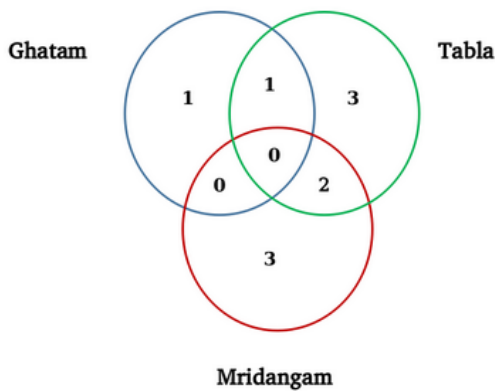
**Tabla** [total: 6] - one out of A and B (condition 1); F and G (condition 3); D (condition 2); H (condition 4 and 5); E {based on the condition given in the question}

Of these experts, A, B and G are experts of Mridangam and D is an expert of Ghatam as well. Thus, excluding these, we have F, H and E who are the experts solely in Tabla. Thus, Option A is the correct answer.

VIDEO SOLUTION

39. B

Based on the given information, we can form the following Venn-diagram for ease of understanding:



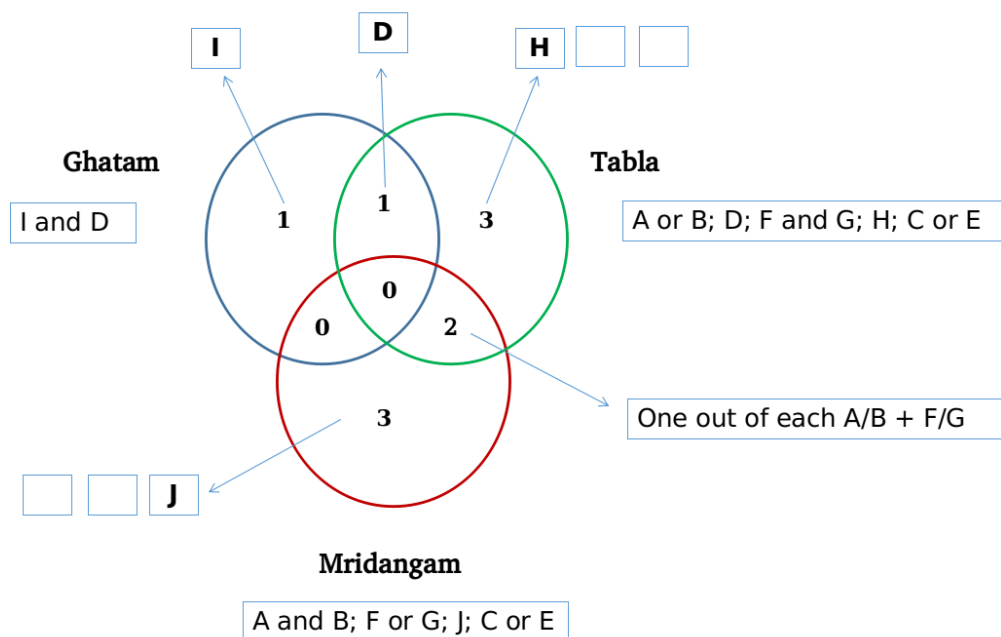
The conditions help us to further bifurcate the individuals based on their expertise.

Mridangam: A and B (condition 1); one out of F and G (condition 3)

Tabla: one out of A and B (condition 1); F and G (condition 3); D (condition 2)

Ghatam: D (condition 2)

Based on condition 4, we infer that 'I' and 'J' are either experts in Ghatam or Mridangam. However, condition 5 adds that 'I' is not an expert in Mridangam. This helps us definitively zero-in on 'I' as an expert in Ghatam. Since 'I' is a Ghatam expert, J is an expert in Mridangam and 'H' is an expert in Tabla [based on conditions 4 and 5]. Thus, we can depict our understanding so far as follows:



**Mridangam** [total: 5] - A and B (condition 1); one out of F and G (condition 3); J (condition 4 and 5); one out of C and E {remaining experts}

**Tabla** [total: 6] - one out of A and B (condition 1); F and G (condition 3); D (condition 2); H (condition 4 and 5); one out of C and E {remaining experts}

**Ghatam** [total: 2] - D (condition 2); I (condition 4 and 5)

Thus, we observe that **J** is definitely an expert in mridangam but not in either tabla or ghatam. Hence, Option B is the correct answer.

VIDEO SOLUTION

#### 40.40

It is given that the total number of products supermarket sells is 320.

cosmetic + nutrition = foreign + domestic = FDA + EU = 320 products

In statement 1, it is given that the number of foreign products is equal to the number of domestic products.

Foreign products = Domestic products =  $320/2 = 160$

|           |                   |           |            |             |
|-----------|-------------------|-----------|------------|-------------|
| Total(32) | Domestic<br>(160) | Cosmetic  | only EU -  | Total EU -  |
|           |                   |           | only FDA - | Total FDA - |
|           |                   |           | Both -     |             |
|           |                   | Nutrition | only EU -  | Total EU -  |
|           |                   |           | only FDA - | Total FDA - |
|           |                   |           | Both -     |             |
|           | Foreign<br>(160)  | Cosmetic  | EU -       |             |
|           |                   |           | FDA -      |             |
|           |                   | Nutrition | EU -       |             |
|           |                   |           | FDA -      |             |

In statement 2, it is given that half of the domestic products were FDA approved cosmetic products, i.e. domestic, cosmetic and FDA = 80

In statement 4, it is given that there were 140 nutrition products, half of them were foreign products. This implies remaining half are domestic.

|           |                   |                   |            |                |
|-----------|-------------------|-------------------|------------|----------------|
| Total(32) | Domestic<br>(160) | Cosmetic<br>(90)  | only EU -  | Total EU -     |
|           |                   |                   | only FDA - | Total FDA - 80 |
|           |                   |                   | Both -     |                |
|           |                   | Nutrition<br>(70) | only EU -  | Total EU -     |
|           |                   |                   | only FDA - | Total FDA -    |
|           |                   |                   | Both -     |                |
|           | Foreign<br>(160)  | Cosmetic<br>(90)  | EU -       |                |
|           |                   |                   | FDA -      |                |
|           |                   | Nutrition<br>(70) | EU -       |                |
|           |                   |                   | FDA -      |                |

In statement 5, it is given that there are 200 FDA approved products out of which 70 are foreign products and 120 are cosmetic products.

If 70 are foreign products, remaining 130 should be domestic products. In domestic products, FDA approved cosmetic products are 80. This implies FDA approved nutrition products are  $130-80$ , i.e. 50.

There are 120 FDA approved cosmetic products.

Domestic, cosmetic and FDA approved = 80

This implies, Foreign, cosmetic and FDA approved is  $120-80$ , i.e. 40.

There are 70 FDA approved foreign products.

This implies Foreign, nutrition and FDA approved is  $70-40$ , i.e. 30.

|           |                   |                   |            |                |
|-----------|-------------------|-------------------|------------|----------------|
| Total(32) | Domestic<br>(160) | Cosmetic<br>(90)  | only EU -  | Total EU -     |
|           |                   |                   | only FDA - | Total FDA - 80 |
|           |                   |                   | Both -     |                |
|           |                   | Nutrition<br>(70) | only EU -  | Total EU -     |
|           |                   |                   | only FDA - | Total FDA - 50 |
|           |                   |                   | Both -     |                |
|           | Foreign<br>(160)  | Cosmetic<br>(90)  | EU - 50    |                |
|           |                   |                   | FDA - 40   |                |
|           |                   | Nutrition<br>(70) | EU - 40    |                |
|           |                   |                   | FDA - 30   |                |

Domestic and Cosmetic = 90

Domestic, cosmetic and FDA approved = 80

This implies, Domestic, cosmetic and FDA not approved is 90-80, i.e. 10.

Therefore, (domestic, cosmetic and only EU) = 10

Similarly, we get (domestic, nutrition and only EU) = 70-50 = 20

|           |                   |                   |              |                |
|-----------|-------------------|-------------------|--------------|----------------|
| Total(32) | Domestic<br>(160) | Cosmetic<br>(90)  | only EU - 10 | Total EU -     |
|           |                   |                   | only FDA -   | Total FDA - 80 |
|           |                   |                   | Both -       |                |
|           |                   | Nutrition<br>(70) | only EU - 20 | Total EU -     |
|           |                   |                   | only FDA -   | Total FDA - 50 |
|           |                   |                   | Both -       |                |
|           | Foreign<br>(160)  | Cosmetic<br>(90)  | EU - 50      |                |
|           |                   |                   | FDA - 40     |                |
|           |                   | Nutrition<br>(70) | EU - 40      |                |
|           |                   |                   | FDA - 30     |                |

The number of foreign, cosmetic and FDA approved products is 40.

The answer is 40.

 VIDEO SOLUTION



# CAT Syllabus PDF



41. D

It is given that the total number of products supermarket sells is 320.

cosmetic + nutrition = foreign + domestic = FDA + EU = 320 products

In statement 1, it is given that the number of foreign products is equal to the number of domestic products.

Foreign products = Domestic products =  $320/2 = 160$



|           |                   |           |            |             |
|-----------|-------------------|-----------|------------|-------------|
| Total(32) | Domestic<br>(160) | Cosmetic  | only EU -  | Total EU -  |
|           |                   |           | only FDA - | Total FDA - |
|           |                   |           | Both -     |             |
|           |                   | Nutrition | only EU -  | Total EU -  |
|           |                   |           | only FDA - | Total FDA - |
|           |                   |           | Both -     |             |
|           | Foreign<br>(160)  | Cosmetic  | EU -       |             |
|           |                   |           | FDA -      |             |
|           |                   | Nutrition | EU -       |             |
|           |                   |           | FDA -      |             |

In statement 2, it is given that half of the domestic products were FDA approved cosmetic products, i.e. domestic, cosmetic and FDA = 80

In statement 4, it is given that there were 140 nutrition products, half of them were foreign products. This implies remaining half are domestic.

|           |                   |                   |            |                |
|-----------|-------------------|-------------------|------------|----------------|
| Total(32) | Domestic<br>(160) | Cosmetic<br>(90)  | only EU -  | Total EU -     |
|           |                   |                   | only FDA - | Total FDA - 80 |
|           |                   |                   | Both -     |                |
|           |                   | Nutrition<br>(70) | only EU -  | Total EU -     |
|           |                   |                   | only FDA - | Total FDA -    |
|           |                   |                   | Both -     |                |
|           | Foreign<br>(160)  | Cosmetic<br>(90)  | EU -       |                |
|           |                   |                   | FDA -      |                |
|           |                   | Nutrition<br>(70) | EU -       |                |
|           |                   |                   | FDA -      |                |

In statement 5, it is given that there are 200 FDA approved products out of which 70 are foreign products and 120 are cosmetic products.

If 70 are foreign products, remaining 130 should be domestic products. In domestic products, FDA approved cosmetic products are 80. This implies FDA approved nutrition products are 130-80, i.e. 50.

There are 120 FDA approved cosmetic products.

Domestic, cosmetic and FDA approved = 80

This implies, Foreign, cosmetic and FDA approved is 120-80, i.e. 40.

There are 70 FDA approved foreign products.

This implies Foreign, nutrition and FDA approved is 70-40, i.e. 30.

|           |                   |                   |            |                |
|-----------|-------------------|-------------------|------------|----------------|
| Total(32) | Domestic<br>(160) | Cosmetic<br>(90)  | only EU -  | Total EU -     |
|           |                   |                   | only FDA - | Total FDA - 80 |
|           |                   |                   | Both -     |                |
|           |                   | Nutrition<br>(70) | only EU -  | Total EU -     |
|           |                   |                   | only FDA - | Total FDA - 50 |
|           |                   |                   | Both -     |                |
|           | Foreign<br>(160)  | Cosmetic<br>(90)  | EU - 50    |                |
|           |                   |                   | FDA - 40   |                |
|           |                   | Nutrition<br>(70) | EU - 40    |                |
|           |                   |                   | FDA - 30   |                |

Domestic and Cosmetic = 90

Domestic, cosmetic and FDA approved = 80

This implies, Domestic, cosmetic and FDA not approved is 90-80, i.e. 10.

Therefore, (domestic, cosmetic and only EU) = 10

Similarly, we get (domestic, nutrition and only EU) =  $70 - 50 = 20$

|           |                   |                   |              |                |
|-----------|-------------------|-------------------|--------------|----------------|
| Total(32) | Domestic<br>(160) | Cosmetic<br>(90)  | only EU - 10 | Total EU -     |
|           |                   |                   | only FDA -   | Total FDA - 80 |
|           |                   |                   | Both -       |                |
|           |                   | Nutrition<br>(70) | only EU - 20 | Total EU -     |
|           |                   |                   | only FDA -   | Total FDA - 50 |
|           |                   |                   | Both -       |                |
|           | Foreign<br>(160)  | Cosmetic<br>(90)  | EU - 50      |                |
|           |                   |                   | FDA - 40     |                |
|           |                   | Nutrition<br>(70) | EU - 40      |                |
|           |                   |                   | FDA - 30     |                |

The number of cosmetic products which do not have FDA approval = domestic only EU + foreign EU =  $10 + 50 = 60$

The answer is option D.

[VIDEO SOLUTION](#)

#### 42. C

It is given that the total number of products supermarket sells is 320.

cosmetic + nutrition = foreign + domestic = FDA + EU = 320 products

In statement 1, it is given that the number of foreign products is equal to the number of domestic products.

Foreign products = Domestic products =  $320/2 = 160$

|           |                   |           |            |             |
|-----------|-------------------|-----------|------------|-------------|
| Total(32) | Domestic<br>(160) | Cosmetic  | only EU -  | Total EU -  |
|           |                   |           | only FDA - | Total FDA - |
|           |                   |           | Both -     |             |
|           |                   | Nutrition | only EU -  | Total EU -  |
|           |                   |           | only FDA - | Total FDA - |
|           |                   |           | Both -     |             |
|           | Foreign<br>(160)  | Cosmetic  | EU -       |             |
|           |                   |           | FDA -      |             |
|           |                   | Nutrition | EU -       |             |
|           |                   |           | FDA -      |             |

In statement 2, it is given that half of the domestic products were FDA approved cosmetic products, i.e. domestic, cosmetic and FDA = 80

In statement 4, it is given that there were 140 nutrition products, half of them were foreign products. This implies remaining half are domestic.

|           |                   |                   |            |                |
|-----------|-------------------|-------------------|------------|----------------|
| Total(32) | Domestic<br>(160) | Cosmetic<br>(90)  | only EU -  | Total EU -     |
|           |                   |                   | only FDA - | Total FDA - 80 |
|           |                   |                   | Both -     |                |
|           |                   | Nutrition<br>(70) | only EU -  | Total EU -     |
|           |                   |                   | only FDA - | Total FDA -    |
|           |                   |                   | Both -     |                |
|           | Foreign<br>(160)  | Cosmetic<br>(90)  | EU -       |                |
|           |                   |                   | FDA -      |                |
|           |                   | Nutrition<br>(70) | EU -       |                |
|           |                   |                   | FDA -      |                |

In statement 5, it is given that there are 200 FDA approved products out of which 70 are foreign products and 120 are cosmetic products.

If 70 are foreign products, remaining 130 should be domestic products. In domestic products, FDA approved

cosmetic products are 80. This implies FDA approved nutrition products are 130-80, i.e. 50.

There are 120 FDA approved cosmetic products.

Domestic, cosmetic and FDA approved = 80

This implies, Foreign, cosmetic and FDA approved is 120-80, i.e. 40.

There are 70 FDA approved foreign products.

This implies Foreign, nutrition and FDA approved is 70-40, i.e. 30.

|           |                   |                   |            |                |
|-----------|-------------------|-------------------|------------|----------------|
| Total(32) | Domestic<br>(160) | Cosmetic<br>(90)  | only EU -  | Total EU -     |
|           |                   |                   | only FDA - | Total FDA - 80 |
|           |                   |                   | Both -     |                |
|           |                   | Nutrition<br>(70) | only EU -  | Total EU -     |
|           |                   |                   | only FDA - | Total FDA - 50 |
|           |                   |                   | Both -     |                |
|           | Foreign<br>(160)  | Cosmetic<br>(90)  | EU - 50    |                |
|           |                   |                   | FDA - 40   |                |
|           |                   | Nutrition<br>(70) | EU - 40    |                |
|           |                   |                   | FDA - 30   |                |

Domestic and Cosmetic = 90

Domestic, cosmetic and FDA approved = 80

This implies, Domestic, cosmetic and FDA not approved is 90-80, i.e. 10.

Therefore, (domestic, cosmetic and only EU) = 10

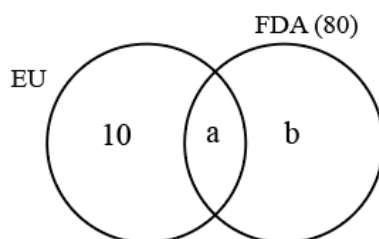
Similarly, we get (domestic, nutrition and only EU) = 70-50 = 20

|           |                   |                   |              |                |
|-----------|-------------------|-------------------|--------------|----------------|
| Total(32) | Domestic<br>(160) | Cosmetic<br>(90)  | only EU - 10 | Total EU -     |
|           |                   |                   | only FDA -   | Total FDA - 80 |
|           |                   |                   | Both -       |                |
|           |                   | Nutrition<br>(70) | only EU - 20 | Total EU -     |
|           |                   |                   | only FDA -   | Total FDA - 50 |
|           |                   |                   | Both -       |                |
|           | Foreign<br>(160)  | Cosmetic<br>(90)  | EU - 50      |                |
|           |                   |                   | FDA - 40     |                |
|           |                   | Nutrition<br>(70) | EU - 40      |                |
|           |                   |                   | FDA - 30     |                |

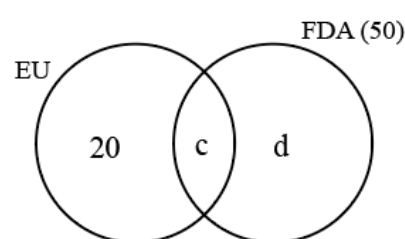
In the question, it is given that 70 cosmetic products did not have EU approval.

In foreign, 40 cosmetic products did not have EU approval. This implies 30 cosmetic products should have only FDA approval in domestic products.

Cosmetic (90)



Nutrition (70)

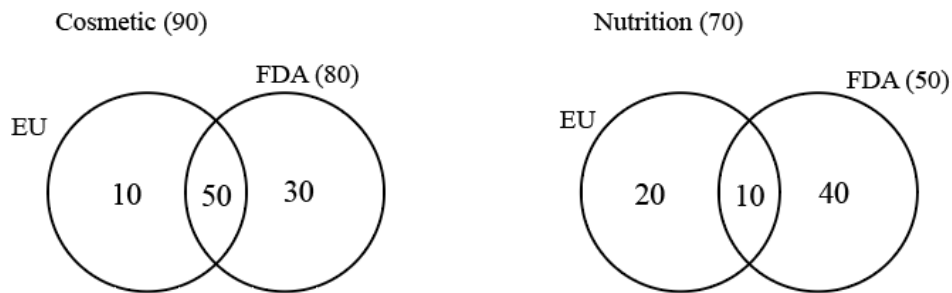


According to the above statement, b = 30

$a = 80 - 30 = 50$

Given,  $a + c = 60$

$c = 60 - 50 = 10$



Therefore, the number of nutrition products which had both the approvals is 10.

The answer is option C.

[▶ VIDEO SOLUTION](#)

**43. 50**

It is given that the total number of products supermarket sells is 320.

cosmetic + nutrition = foreign + domestic = FDA + EU = 320 products

In statement 1, it is given that the number of foreign products is equal to the number of domestic products.

Foreign products = Domestic products =  $320/2 = 160$

|           |                   |           |            |             |
|-----------|-------------------|-----------|------------|-------------|
| Total(32) | Domestic<br>(160) | Cosmetic  | only EU -  | Total EU -  |
|           |                   |           | only FDA - | Total FDA - |
|           |                   |           | Both -     |             |
|           |                   | Nutrition | only EU -  | Total EU -  |
|           |                   |           | only FDA - | Total FDA - |
|           |                   |           | Both -     |             |
|           | Foreign<br>(160)  | Cosmetic  | EU -       |             |
|           |                   |           | FDA -      |             |
|           |                   | Nutrition | EU -       |             |
|           |                   |           | FDA -      |             |

In statement 2, it is given that half of the domestic products were FDA approved cosmetic products, i.e. domestic, cosmetic and FDA = 80

In statement 4, it is given that there were 140 nutrition products, half of them were foreign products. This implies remaining half are domestic.

|           |                   |                   |            |                |
|-----------|-------------------|-------------------|------------|----------------|
| Total(32) | Domestic<br>(160) | Cosmetic<br>(90)  | only EU -  | Total EU -     |
|           |                   |                   | only FDA - | Total FDA - 80 |
|           |                   |                   | Both -     |                |
|           |                   | Nutrition<br>(70) | only EU -  | Total EU -     |
|           |                   |                   | only FDA - | Total FDA -    |
|           |                   |                   | Both -     |                |
|           | Foreign<br>(160)  | Cosmetic<br>(90)  | EU -       |                |
|           |                   |                   | FDA -      |                |
|           |                   | Nutrition<br>(70) | EU -       |                |
|           |                   |                   | FDA -      |                |

In statement 5, it is given that there are 200 FDA approved products out of which 70 are foreign products and 120 are cosmetic products.

If 70 are foreign products, remaining 130 should be domestic products. In domestic products, FDA approved cosmetic products are 80. This implies FDA approved nutrition products are  $130-80$ , i.e. 50.

There are 120 FDA approved cosmetic products.

Domestic, cosmetic and FDA approved = 80

This implies, Foreign, cosmetic and FDA approved is  $120-80$ , i.e. 40.

This implies Foreign, nutrition and FDA approved is 70-40, i.e. 30.

|                   |                   |                   |            |                |
|-------------------|-------------------|-------------------|------------|----------------|
| Total(32)         | Domestic<br>(160) | Cosmetic<br>(90)  | only EU -  | Total EU -     |
|                   |                   |                   | only FDA - | Total FDA - 80 |
|                   |                   |                   | Both -     |                |
|                   |                   | Nutrition<br>(70) | only EU -  | Total EU -     |
|                   |                   |                   | only FDA - | Total FDA - 50 |
|                   |                   |                   | Both -     |                |
|                   | Foreign<br>(160)  | Cosmetic<br>(90)  | EU - 50    |                |
|                   |                   |                   | FDA - 40   |                |
| Nutrition<br>(70) |                   | EU - 40           |            |                |
|                   |                   | FDA - 30          |            |                |

Domestic, comestic and FDA approved = 80

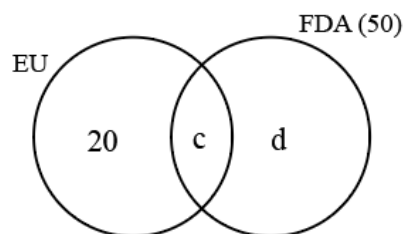
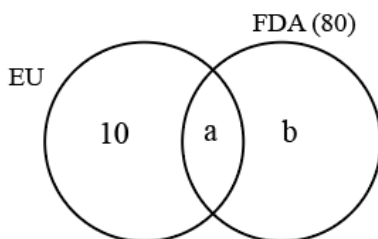
Therefore, (domestic, cosmetic and only EU) = 10

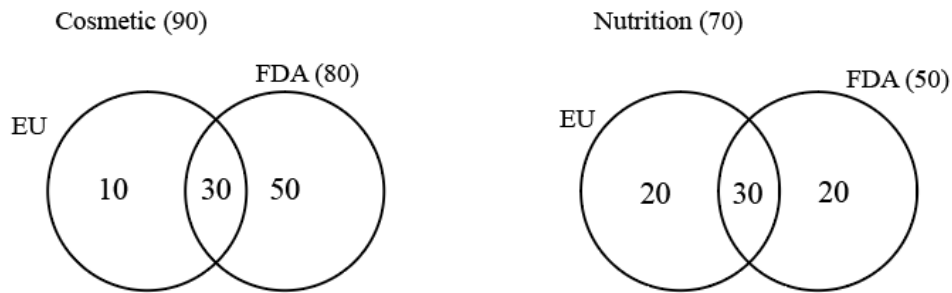
Similarly, we get (domestic, nutrition and only EU) =  $70 - 50 = 20$

|                   |                   |                   |              |                |
|-------------------|-------------------|-------------------|--------------|----------------|
| Total(32)         | Domestic<br>(160) | Cosmetic<br>(90)  | only EU - 10 | Total EU -     |
|                   |                   |                   | only FDA -   | Total FDA - 80 |
|                   |                   |                   | Both -       |                |
|                   |                   | Nutrition<br>(70) | only EU - 20 | Total EU -     |
|                   |                   |                   | only FDA -   | Total FDA - 50 |
|                   |                   |                   | Both -       |                |
|                   | Foreign<br>(160)  | Cosmetic<br>(90)  | EU - 50      |                |
|                   |                   |                   | FDA - 40     |                |
| Nutrition<br>(70) |                   | EU - 40           |              |                |
|                   |                   | FDA - 30          |              |                |

In Foreign products, there are 30 nutrition products which do not have EU approval. This implies 20 nutrition products do not have EU(have only FDA) approval in domestic products.

Nutrition (70)


$$c = 50 - 20 = 30$$
$$a = 60 - 30 = 30$$
$$b = 80 - 30 = 50$$



Therefore, the number of domestic cosmetic products did not have EU(only FDA) approval is 50.

[VIDEO SOLUTION](#)

**44. A**

It is given that the total number of products supermarket sells is 320.

cosmetic + nutrition = foreign + domestic = FDA + EU = 320 products

In statement 1, it is given that the number of foreign products is equal to the number of domestic products.

Foreign products = Domestic products =  $320/2 = 160$

|           |                   |           |            |             |
|-----------|-------------------|-----------|------------|-------------|
| Total(32) | Domestic<br>(160) | Cosmetic  | only EU -  | Total EU -  |
|           |                   |           | only FDA - | Total FDA - |
|           |                   |           | Both -     |             |
|           |                   | Nutrition | only EU -  | Total EU -  |
|           |                   |           | only FDA - | Total FDA - |
|           |                   |           | Both -     |             |
|           | Foreign<br>(160)  | Cosmetic  | EU -       |             |
|           |                   |           | FDA -      |             |
|           |                   | Nutrition | EU -       |             |
|           |                   |           | FDA -      |             |

In statement 2, it is given that half of the domestic products were FDA approved cosmetic products, i.e. domestic, cosmetic and FDA = 80

In statement 4, it is given that there were 140 nutrition products, half of them were foreign products. This implies remaining half are domestic.

|           |                   |                   |            |                |
|-----------|-------------------|-------------------|------------|----------------|
| Total(32) | Domestic<br>(160) | Cosmetic<br>(90)  | only EU -  | Total EU -     |
|           |                   |                   | only FDA - | Total FDA - 80 |
|           |                   |                   | Both -     |                |
|           |                   | Nutrition<br>(70) | only EU -  | Total EU -     |
|           |                   |                   | only FDA - | Total FDA -    |
|           |                   |                   | Both -     |                |
|           | Foreign<br>(160)  | Cosmetic<br>(90)  | EU -       |                |
|           |                   |                   | FDA -      |                |
|           |                   | Nutrition<br>(70) | EU -       |                |
|           |                   |                   | FDA -      |                |

In statement 5, it is given that there are 200 FDA approved products out of which 70 are foreign products and 120 are cosmetic products.

If 70 are foreign products, remaining 130 should be domestic products. In domestic products, FDA approved cosmetic products are 80. This implies FDA approved nutrition products are  $130-80$ , i.e. 50.

There are 120 FDA approved cosmetic products.

Domestic, cosmetic and FDA approved = 80

This implies, Foreign, cosmetic and FDA approved is  $120-80$ , i.e. 40.

There are 70 FDA approved foreign products.

This implies Foreign, nutrition and FDA approved is 70-40, i.e. 30.

|           |                   |                   |            |                |
|-----------|-------------------|-------------------|------------|----------------|
| Total(32) | Domestic<br>(160) | Cosmetic<br>(90)  | only EU -  | Total EU -     |
|           |                   |                   | only FDA - | Total FDA - 80 |
|           |                   |                   | Both -     |                |
|           |                   | Nutrition<br>(70) | only EU -  | Total EU -     |
|           |                   |                   | only FDA - | Total FDA - 50 |
|           |                   |                   | Both -     |                |
|           | Foreign<br>(160)  | Cosmetic<br>(90)  | EU - 50    |                |
|           |                   |                   | FDA - 40   |                |
|           |                   | Nutrition<br>(70) | EU - 40    |                |
|           |                   |                   | FDA - 30   |                |

Domestic and Cosmetic = 90

Domestic, cosmetic and FDA approved = 80

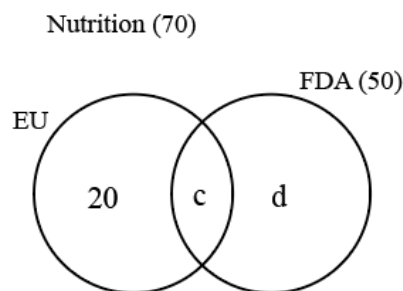
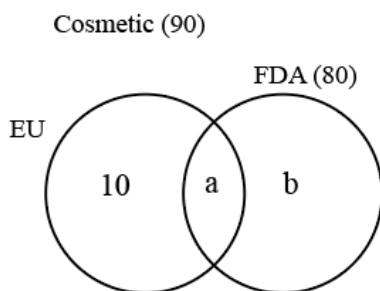
This implies, Domestic, cosmetic and FDA not approved is 90-80, i.e. 10.

Therefore, (domestic, cosmetic and only EU) = 10

Similarly, we get (domestic, nutrition and only EU) = 70-50 = 20

|           |                   |                   |              |                |
|-----------|-------------------|-------------------|--------------|----------------|
| Total(32) | Domestic<br>(160) | Cosmetic<br>(90)  | only EU - 10 | Total EU -     |
|           |                   |                   | only FDA -   | Total FDA - 80 |
|           |                   |                   | Both -       |                |
|           |                   | Nutrition<br>(70) | only EU - 20 | Total EU -     |
|           |                   |                   | only FDA -   | Total FDA - 50 |
|           |                   |                   | Both -       |                |
|           | Foreign<br>(160)  | Cosmetic<br>(90)  | EU - 50      |                |
|           |                   |                   | FDA - 40     |                |
|           |                   | Nutrition<br>(70) | EU - 40      |                |
|           |                   |                   | FDA - 30     |                |

In statement 3, it is given that the number of domestic products which have both the approvals = 60



In the question, it is given that  $a + c = 60$

To find the minimum value of  $a$ , we need to maximise  $c$ .

Maximum value  $c$  can take is 50

Therefore, minimum value of  $a$  is 60-50, i.e. 10.

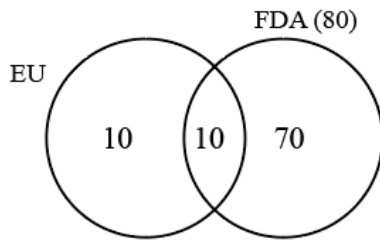
To find the maximum value of  $a$ , we need to minimise  $c$ .

Maximum value  $c$  can take is 0.

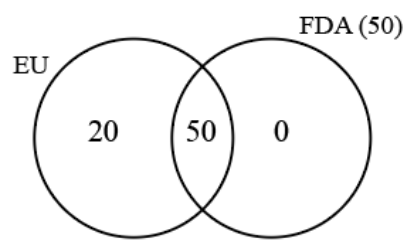
Therefore, maximum value of  $a$  is 60-0, i.e. 60.

$a$  is minimum:

Cosmetic (90)



Nutrition (70)

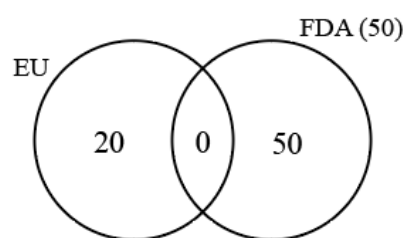


a is maximum:

Cosmetic (90)



Nutrition (70)



Therefore, the number of domestic cosmetic products that had both the approvals is at least 10 and at most 60.

The answer is option A.

[VIDEO SOLUTION](#)

45.20

Let the number of schools with exactly three of the facilities was the same irrespective of which three were considered be  $x$ .

Number of schools with none of the facilities be 'n' from 1,  $n=80$ .

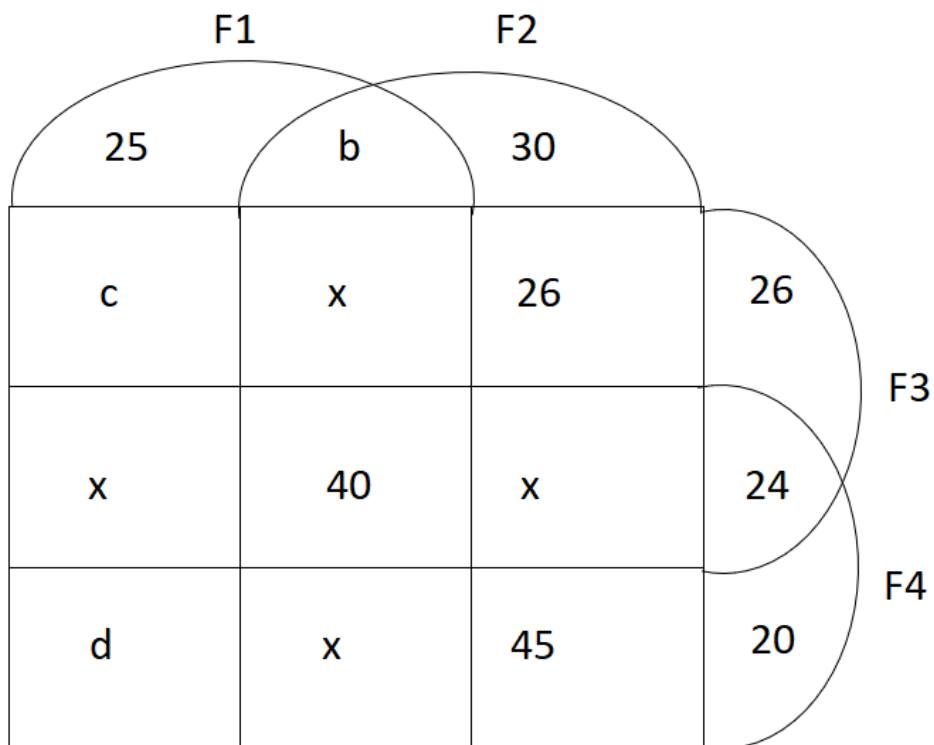
Number of schools with only F1 and F2 be 'b'

Number of schools with only F1 and F3 be 'c'

Number of schools with only F1 and F4 be 'd'

From the information given in the question we will get the following Venn diagram.





From 5,  $b+141+3x=313 \Rightarrow b+3x=172 \dots (i)$

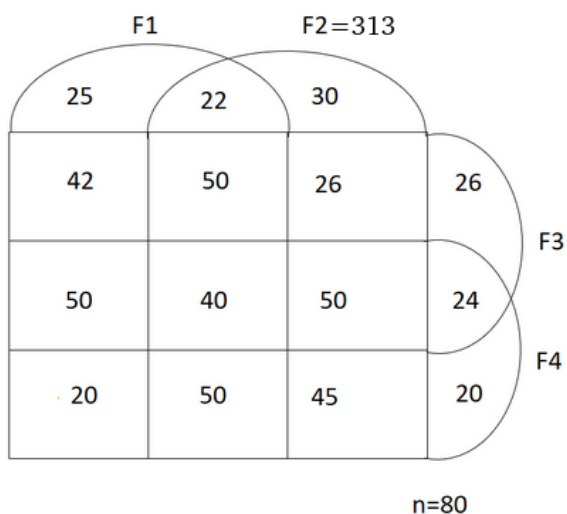
From 8,  $b+x+40+x=162 \Rightarrow b+2x=122 \dots (ii)$

$(ii)-(i)$  gives  $x=50 \Rightarrow b=22$

From 9,  $237+3x+c+d=279+3x=d \Rightarrow c=42$

Total number of schools = 600  $\Rightarrow 313+25+c+x+d+26+24+20+80=600 \Rightarrow d=20$ .

The final table looks like:



The total number of schools having only F1 and F4=20.

[VIDEO SOLUTION](#)



46. A

Let the number of schools with exactly three of the facilities was the same irrespective of which three were considered be  $x$ .

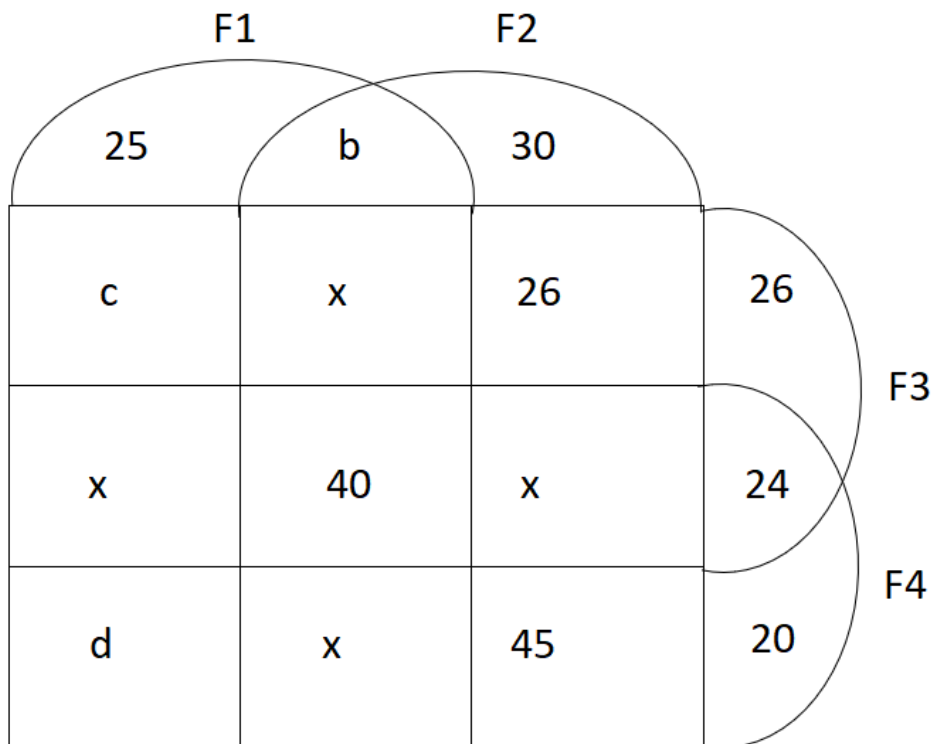
Number of schools with none of the facilities be 'n' from 1,  $n=80$ .

Number of schools with only F1 and F2 be 'b'

Number of schools with only F1 and F3 be 'c'

Number of schools with only F1 and F4 be 'd'

From the information given in the question we will get the following Venn diagram.



From 5,  $b+141+3x=313 \Rightarrow b+3x=172 \dots (i)$

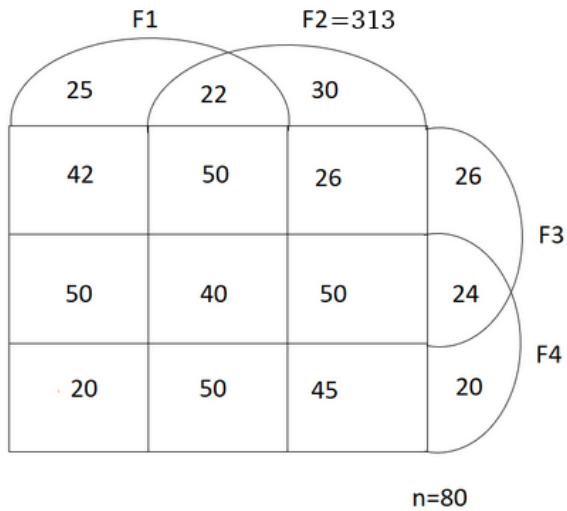
From 8,  $b+x+40+x=162 \Rightarrow b+2x=122 \dots (ii)$

(ii)-(i) gives  $x=50 \Rightarrow b=22$

From 9,  $237+3x+c+d=279+3x=d \Rightarrow c=42$

Total number of schools = 600  $\Rightarrow 313+25+c+x+d+26+24+20+80=600 \Rightarrow d=20$ .

The final table looks like:



The total number of schools having facilities F4 and F2=  $45+50+50+40=185$ .

[VIDEO SOLUTION](#)

47. C

Let the number of schools with exactly three of the facilities was the same irrespective of which three were considered be  $x$ .

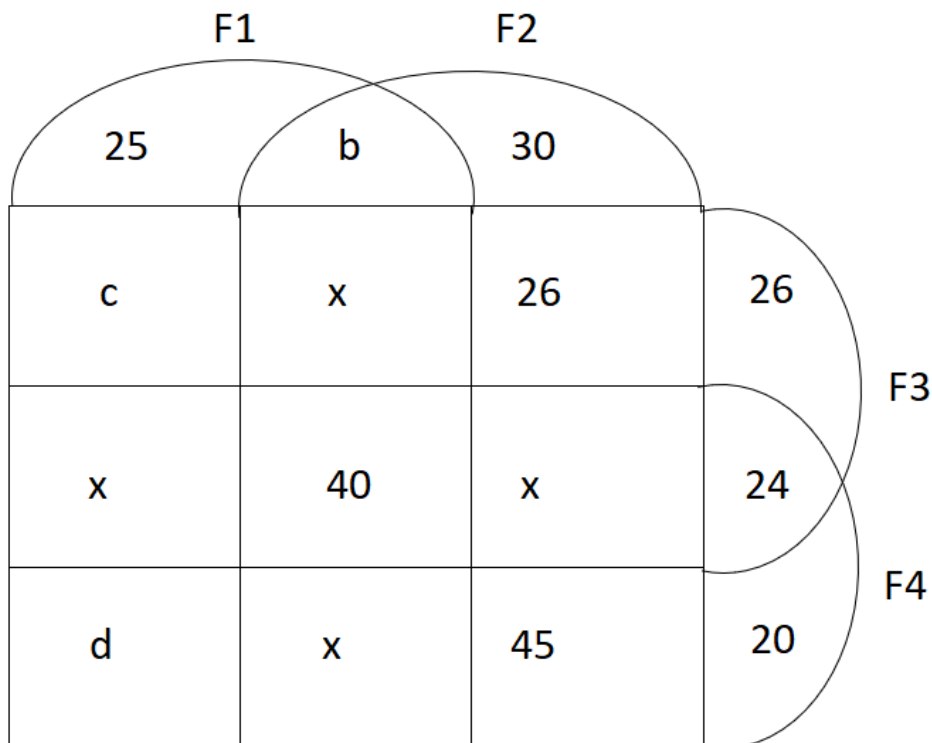
Number of schools with none of the facilities be 'n' from 1,  $n=80$ .

Number of schools with only F1 and F2 be 'b'

Number of schools with only F1 and F3 be 'c'

Number of schools with only F1 and F4 be 'd'

From the information given in the question we will get the following Venn diagram.



From 5,  $b+141+3x=313 \Rightarrow b+3x=172 \dots (i)$

From 8,  $b+x+40+x=162 \Rightarrow b+2x=122 \dots (ii)$

(ii)-(i) gives  $x=50 \Rightarrow b=22$

From 9,  $237+3x+c+d=279+3x=d \Rightarrow c=42$

Total number of schools = 600  $\Rightarrow 313+25+c+x+d+26+24+20+80=600 \Rightarrow d=20$ .

The final table looks like:

| F1   |    | F2=313 |    |    |
|------|----|--------|----|----|
| 25   | 22 | 30     |    |    |
| 42   | 50 | 26     | 26 |    |
| 50   | 40 | 50     | 24 | F3 |
| 20   | 50 | 45     | 20 | F4 |
| n=80 |    |        |    |    |

The total number of schools with exactly three of the four facilities =  $4x=200$ .

[VIDEO SOLUTION](#)

#### 48.42

Let the number of schools with exactly three of the facilities was the same irrespective of which three were considered be  $x$ .

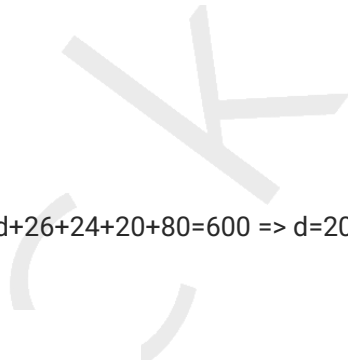
Number of schools with none of the facilities be ' $n$ ' from 1,  $n=80$ .

Number of schools with only F1 and F2 be ' $b$ '

Number of schools with only F1 and F3 be ' $c$ '

Number of schools with only F1 and F4 be ' $d$ '

From the information given in the question we will get the following Venn diagram.

$$d+26+24+20+80=600 \Rightarrow d=20$$
$$d+26+24+20+80=600 \Rightarrow d=20$$
$$d+26+24+20+80=600 \Rightarrow d=20$$
$$d+26+24+20+80=600 \Rightarrow d=20$$
$$d+26+24+20+80=600 \Rightarrow d=20$$
$$d+26+24+20+80=600 \Rightarrow d=20$$

$$d+26+24+20+80=600 \Rightarrow d=20$$
$$d+26+24+20+80=600 \Rightarrow d=20$$